



Chapter # 04

Urinary System of Man



Student Learning Outcomes

At the end of this chapter, students will be able to:

- [B-12-R-09] List various nitrogenous compounds excreted during the process of excretion.
- [B-12-R-10] Explain the nature of excretory products in relation to habitats.
- [B-12-R-11] Outline different organs of the urinary system.
- [B-12-R-12] Describe the structure of the kidney.
- [B-12-R-13] Relate the structure of kidney with its function.
- [B-12-R-14] Explain the detailed structure of a nephron.
- [B-12-R-15] Explain the process of glomerular filtration, selective re-absorption and tubular secretion as the events in kidney functioning.
- [B-12-R-16] Explain regulatory mechanism involved in concentration of urine.
- [B-12-R-17] Justify the functioning of kidneys as both excretion and osmoregulation.
- [B-12-R-18] Compare the function of two major capillary beds in kidneys i.e., glomerular capillaries and peritubular capillaries.
- [B-12-R-19] List the urinary tract infections and the bacteria responsible.
- [B-12-R-20] Explain the causes and treatments of kidney stones.
- [B-12-R-21] Outline the causes of kidney failure.
- [B-12-R-22] Explain in detail the mechanism and problems related to dialysis.
- [B-12-R-23] Describe the principles and the problems associated with kidney transplant.

Metabolic activities within body cells generate a range of waste products, many of which can become harmful if allowed to accumulate. To prevent toxicity, these wastes must be efficiently removed. Body fluids such as blood and lymph serve as transport mediums, carrying waste materials away from active tissues to excretory organs.

Different systems play distinct roles in excretion. The **respiratory system** eliminates carbon dioxide, a major waste product of cellular respiration, from the bloodstream. The **urinary system** is responsible for filtering blood to remove nitrogenous wastes (such as urea), excess salts, and water.

In addition to excretion, the urinary system performs several vital regulatory functions. It maintains the osmotic balance of body fluids by adjusting water and electrolyte concentrations, regulates blood pH and fluid volume, and contributes to the control of **blood pressure** and **red blood cell production** through the secretion of hormones like **renin** and **erythropoietin**.

SLO [B-12-R-09]

List various nitrogenous compounds excreted during the process of excretion.

NITROGENOUS COMPOUNDS EXCRETED DURING EXCRETION

The breakdown of proteins and nucleic acids in the body produces nitrogenous wastes. These substances are toxic if allowed to accumulate, so they must be removed efficiently through the process of excretion. The main nitrogenous compounds excreted by humans include urea, uric acid, creatinine, and small amounts of ammonia. Each of these compounds originates from different metabolic processes and is eliminated mainly through urine.



1. Urea

Urea is the **primary nitrogenous waste** in humans. It is formed in the **liver** through the urea cycle, where ammonia (a highly toxic by-product of protein breakdown) is converted into a less toxic, water-soluble form. Urea is then transported by the blood to the kidneys and excreted in urine. On average, about **80–90% of the nitrogen excreted** from the body is in the form of urea. For example, after eating a protein-rich meal, urea levels in the blood and urine rise because more amino acids are broken down.

2. Uric Acid

Uric acid is produced during the **breakdown of nucleic acids (DNA and RNA)**. It is less soluble in water compared to urea. Normally, only a small quantity of uric acid is excreted in urine, but if its concentration becomes too high, it can crystallize and deposit in joints, causing a painful disorder called **gout**. Birds and reptiles excrete uric acid as their main nitrogenous waste, which conserves water since it is eliminated as a paste-like substance.

3. Creatinine

Creatinine is formed in muscles as a by-product of the breakdown of **creatine phosphate**, a compound that stores energy for muscle contractions. It is produced at a relatively constant rate, depending on muscle activity and mass. Because the kidneys efficiently filter creatinine from the blood, measuring its level in urine or blood is an important **clinical test of kidney function**. Elevated creatinine levels may indicate impaired kidney performance.

4. Ammonia

Ammonia is a direct by-product of amino acid metabolism. It is extremely toxic and must be removed quickly from the body. In humans, most ammonia is converted into urea in the liver before excretion. However, small amounts of free ammonia are excreted directly into urine, where they also help *maintain the acid-base balance* of the body.

5. Other Nitrogenous Compounds

In addition to the main wastes, trace amounts of other nitrogen-containing substances such as **amino acids, peptides, and nitrogenous bases** may also appear in urine. Their presence in significant amounts, however, can be a sign of metabolic or renal disorders. For instance, excessive amino acids in urine (aminoaciduria) may indicate defects in amino acid metabolism.

Doctors often perform a **urinalysis** to measure nitrogenous waste levels. For example, the **blood urea nitrogen (BUN)** and **creatinine test** are routinely used to evaluate kidney health. Elevated levels of urea or creatinine in blood may signal kidney disease, dehydration, or other metabolic problems. The nitrogenous compounds excreted during excretion include **urea (major form), uric acid, creatinine, and small amounts of ammonia**. Together, these wastes help the body maintain chemical balance and prevent toxic buildup.

SLO [B-12-R-10]

Explain the nature of excretory products in relation to habitats.

EXCRETORY PRODUCTS AND THEIR RELATION TO HABITATS

Different animals excrete nitrogenous wastes in forms that reflect both their metabolism and the availability of water in their environment. The three main types of excretory products are **ammonia, urea, and uric acid**, and each type is associated with particular habitats and survival strategies. This adaptation ensures that organisms can efficiently remove toxic wastes while conserving water and energy.

1. Ammonotelic Animals: Ammonia Excretion

Ammonia is the most primitive and highly toxic nitrogenous waste. Because it is very soluble in water, it must be *excreted quickly and in a diluted form*. Therefore, animals that live in aquatic habitats, where water is abundantly available, usually excrete ammonia. To safely excrete **just one gram of nitrogen as ammonia**, an organism requires approximately **500 mL of water**, making this strategy feasible only for organisms in water-rich environments.

- **Examples:** Many freshwater invertebrates and vertebrates, such as protozoans, sponges, cnidarians, echinoderms, crustaceans, and most bony fishes, amphibians, and aquatic animals, adopt this method of excretion. These organisms are termed ammonotelic.



- **Advantage:** Ammonia can be released directly into surrounding water through gills or body surfaces without requiring much energy.
- **Limitation:** It requires a continuous supply of water for dilution, which is why this mode is not suitable for terrestrial habitats.

2. Ureotelic Animals: Urea Excretion

Urea is less toxic than ammonia and can be stored in the body for a short time before being excreted. It is **approximately 100,000 times less toxic** than ammonia. It is highly soluble in water and requires less water for excretion compared to ammonia, making it suitable for land-dwelling animals. Urea is synthesized in the liver through the **urea cycle** and excreted mainly through urine.

- **Examples:** Humans and most mammals (e.g., dogs, lions, cows) are ureotelic.
- **Advantage:** Reduces water loss while maintaining effective waste removal.
- **Real-life link:** Desert mammals like the **kangaroo rat** concentrate their urine to conserve maximum water while excreting urea.

3. Uricotelic Animals: Uric Acid Excretion

Uric acid is the least toxic form of nitrogenous waste and is excreted as a paste or crystals, requiring very little water. This adaptation is highly beneficial in arid environments, where water is scarce, and for animals that develop within shelled eggs, where nitrogenous wastes cannot be excreted in liquid form.

- **Examples:** Birds, reptiles, and insects are uricotelic.
- **Advantage:** Saves water and prevents accumulation of soluble toxic waste inside the egg.
- **Real-life link:** The **white paste** often seen in bird droppings is uric acid.

► Comparative Adaptation to Habitats

- **Aquatic habitats:** Favor ammonotelism, as water is available to dilute ammonia.
- **Terrestrial habitats with moderate water:** Favor ureotelism, as animals can afford some water loss but must regulate it.
- **Arid habitats or egg-laying species:** Favor uricotelism, as water conservation is critical.

Type of Excretory Product	Toxicity	Water Requirement	Energy Cost	Examples	Habitat Adaptation
Ammonia (Ammonotelic)	Very high	Very high	Low	Fish, amphibians	Aquatic
Urea (Ureotelic)	Moderate	Moderate	Moderate	Mammals	Terrestrial (with water access)
Uric acid (Uricotelic)	Very low	Very low	High	Birds, reptiles, insects	Arid / Egg-laying

The form of nitrogenous waste excreted by an organism (ammonia, urea, or uric acid) is closely related to its **habitat and water availability**. Aquatic animals are ammonotelic, terrestrial mammals are ureotelic, while birds, reptiles, and insects are uricotelic.

Relationship between Excretory Products and Habitats

The nature of nitrogenous waste excreted by an organism is closely linked to its **habitat and the availability of water** in its environment. Since nitrogenous wastes vary in toxicity and water solubility, different organisms have evolved excretory strategies that reflect their ecological conditions.

- **Ammonia** – excreted primarily by *aquatic* organisms.
- **Urea** – excreted by both *aquatic* and *terrestrial* organisms.
- **Uric acid** – excreted mainly by *terrestrial* organisms.

► Ammonia

Ammonia is the most toxic of all nitrogenous wastes. It readily increases the pH of body fluids and disrupts membrane transport systems, posing an immediate threat to cellular function. However, it is highly water-soluble and diffuses rapidly across biological membranes, allowing for quick elimination—*provided water is abundantly available*.

► Urea

In environments where water is more limited, especially on land or in marine conditions where conserving

SCIENCE TITBITS

Humans excrete small quantities of uric acid but this is produced from the breakdown of nucleic acid and not from breakdown of proteins. Approximately one gram of uric acid is excreted in urine per day.



water is essential, organisms convert ammonia into urea—a compound that is about 100,000 times less toxic than ammonia. Although this conversion requires additional energy, it allows for safer storage and delayed excretion.

Excreting one gram of nitrogen as urea requires only 50 mL of water, making it a more efficient strategy for water conservation.

Organisms that primarily excrete urea are called **ureotelic**. This group includes **mammals** (e.g., humans), **amphibians**, and certain **marine species** such as sharks.

CRITICAL THINKING

Q. Where do you think the carbon dioxide used in the formation of urea comes from? Where does the remainder of excess carbon dioxide go to be excreted?

Ans. The carbon dioxide involved in urea formation comes from **cellular respiration**. Inside the mitochondria of body cells, carbohydrates, fats, and proteins are broken down to produce energy (ATP). A major by-product of this process is **carbon dioxide (CO₂)**.

In the **urea cycle** (which mainly takes place in the liver), some of this CO₂ combines with ammonia (NH₃) to form carbamoyl phosphate, which then enters the cycle and eventually contributes to the production of urea. Urea is later excreted in urine.

The remainder of the excess CO₂, which is not used in the urea cycle, does not stay in the liver. Instead, it dissolves in the blood, is carried to the lungs (mainly as **bicarbonate ions, HCO₃⁻**), and is exhaled during breathing.

SLOs Based (MCQs)

EXCRETION, RELATIONSHIP BETWEEN EXCRETORY PRODUCTS AND HABITATS

- Which nitrogenous waste product is the most toxic and requires the greatest amount of water for excretion?
A. Urea B. Uric acid
C. Ammonia D. Creatinine
- Which of the following organisms is most likely to be ammonotelic?
A. Human B. Frog
C. Bony fish D. Bird
- Which excretory product is most suitable for organisms living in arid conditions?
A. Ammonia B. Urea
C. Uric acid D. Nitric oxide
- Which of the following statements correctly distinguishes excretion from egestion?
A. Excretion eliminates undigested food
B. Excretion removes digestive juices
C. Excretion eliminates metabolic waste; egestion removes undigested food
D. Both refer to removal of waste via the anus
- What is the function of erythropoietin secreted by the kidneys?
A. Regulates glucose levels
B. Enhances breathing rate
C. Stimulates red blood cell production
D. Filters nitrogenous waste
- Which of the following best describes ureotelic animals?
A. Excrete urea in water-rich habitats
B. Use large amounts of water for ammonia removal
C. Excrete urea in moderate water availability
D. Store nitrogen as uric acid in solid form
- How much water is required to excrete one gram of nitrogen as urea?
A. 5 mL B. 50 mL
C. 150 mL D. 500 mL
- Which of the following nitrogenous wastes is produced during the breakdown of nucleic acids in humans?
A. Ammonia B. Urea
C. Creatinine D. Uric acid
- Which waste is removed from the body through the respiratory system?
A. Ammonia B. Urea
C. Carbon dioxide D. Uric acid
- What is the main function of renin released by the kidneys?
A. Dilates bronchi
B. Regulates blood pressure
C. Increases uric acid excretion
D. Neutralizes stomach acid

Answer Key

1. C	2. C	3. C	4. C	5. C	6. C	7. B	8. D	9. C	10. B
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Q.1 What is excretion and how is it different from egestion?

Ans. Excretion is the process of eliminating **metabolic wastes** such as urea, carbon dioxide, and ammonia from the body. It maintains the internal chemical balance and prevents the toxic buildup of waste products.

Egestion, however, is the removal of **undigested and unabsorbed food** from the digestive tract in the form of feces. It is not related to cellular metabolism.

Thus, excretion deals with **metabolic waste products**, while egestion involves **undigested food residues**.

MARKING SCHEME (2 Marks)

- Definition of both terms (1+1), difference (1).

Q.2 Why must nitrogenous wastes be eliminated from the body?

Ans. Nitrogenous wastes such as ammonia, urea, and uric acid arise from the breakdown of proteins and nucleic acids. If allowed to accumulate, they can:

1. **Prove toxic to cells** by interfering with biochemical functions.
2. **Disturb pH balance**, making body fluids either too acidic or alkaline.
3. **Disrupt enzyme activity and membrane function**, impairing normal physiology.
4. Therefore, elimination of nitrogenous wastes is essential to maintain **homeostasis** and ensure the stable functioning of the body.

MARKING SCHEME (3 Marks)

- 1 mark each for three reasons.

Q.3 What are ammonotelic animals? Give two examples.

Ans. Ammonotelic animals are organisms that excrete nitrogen mainly in the form of **ammonia**, a highly toxic but water-soluble compound. Since ammonia can diffuse directly into water, this strategy is effective for aquatic organisms where water is abundantly available to dilute and eliminate the waste.

Examples of ammonotelic animals include **protozoans** and **most bony fishes**.

MARKING SCHEME (3 Marks)

- Definition (1), detoxification mode (1), examples (1).

Q.4 Why is urea a preferred excretory product for terrestrial animals?

Ans. Urea is the major nitrogenous waste in mammals and many terrestrial animals because:

1. It is **less toxic** than ammonia and can be stored temporarily in body fluids.
2. It requires **less water** for elimination—only about 50 mL per gram of nitrogen—helping in water conservation.
3. Although its synthesis in the liver requires energy, this trade-off allows **safe and efficient waste removal** in environments where water is limited.
4. Thus, urea formation is particularly suited for terrestrial habitats.

MARKING SCHEME (3 Marks)

- Three reasons, 1 mark each.

Q.5 Describe the role of the urinary system in homeostasis.

Ans. The urinary system plays a central role in maintaining **homeostasis**. Its kidneys:

1. **Filter blood** to remove nitrogenous wastes, excess salts, and water.
2. Regulate **osmotic balance, pH, and fluid volume** to keep the internal environment stable.
3. Secrete hormones such as **renin**, which helps control blood pressure, and **erythropoietin**, which stimulates red blood cell production.
4. In this way, the urinary system ensures both waste elimination and physiological regulation.

MARKING SCHEME (3 Marks)

- Three roles, 1 mark each.

Q.6 What is the significance of uric acid excretion in birds and insects?

Ans. Uric acid is almost **insoluble in water** and is excreted as a semi-solid paste or crystals. This property allows maximum **conservation of water**, making it highly efficient in dry habitats. For birds, reptiles, and insects, uric acid excretion **minimizes water loss** and enables survival in **arid environments** where water is scarce.

MARKING SCHEME (3 Marks)

- Properties of uric acid (2), advantage to birds and insects (1).

Q.7 How are excretory products related to an organism's habitat?

Ans. The type of nitrogenous waste produced by an organism depends largely on its **habitat and water availability**:

- **Ammonotelic animals** (e.g., protozoans, most bony fishes) excrete ammonia, suitable for aquatic habitats where water is abundant.
- **Ureotelic animals** (e.g., mammals, amphibians) excrete urea, which conserves water compared to ammonia.
- **Uricotelic animals** (e.g., birds, reptiles, insects) excrete uric acid, minimizing water loss in dry environments.

Thus, excretory products are closely adapted to the **ecological conditions** of each organism.

MARKING SCHEME (3 Marks)

- Relation to environment (1), types with examples (2).

10
10

Q.8 Why is ammonia not a suitable excretory product for land animals?

Ans. Ammonia is **highly toxic** and requires about **500 mL of water per gram of nitrogen** to be safely excreted. Such large water requirements are impractical for terrestrial animals that often face **limited water availability**. Therefore, land animals detoxify ammonia by converting it into **urea or uric acid**, which are less toxic and require less water for elimination.

MARKING SCHEME (3 Marks)

- Terrestrial conditions (2), requirement of ammonia excretion (1).

Q.9 What are the main nitrogenous waste products in humans?

Ans. Humans excrete three main nitrogenous wastes:

1. **Urea**—the chief excretory product, formed from amino acid breakdown.
2. **Uric acid**—produced in small amounts from nucleic acid metabolism.
3. **Creatinine**—generated from muscle metabolism.
4. All these products are eliminated through the urine by the kidneys.

MARKING SCHEME (3 Marks)

- 1 mark each for three waste products.

Q.10 Why does the conversion of ammonia to urea require energy?

Ans. Ammonia is produced directly during **protein metabolism**. To reduce its toxicity and water demand, it is converted into **urea** in the liver via the urea cycle. This biochemical process involves **several enzymatic reactions that consume ATP**. Although energy is required, the conversion allows safer storage and efficient excretion with reduced water loss—an essential adaptation for terrestrial animals.

MARKING SCHEME (3 Marks)

- Ammonia production (1), ATP requirement explained (2).

SLO [B-12-R-11]

Outline different organs of the urinary system.

ORGANS OF THE URINARY SYSTEM

The urinary system plays a central role in **excretion and osmoregulation**, ensuring removal of **nitrogenous wastes** while maintaining water, electrolyte, and acid–base balance. The system consists of specialized organs that work in coordination to filter blood, form urine, and eliminate it from the body. The major organs are: **kidneys, ureters, urinary bladder, and urethra**.

1. Kidneys – The Main Excretory Organs

The kidneys are two bean-shaped organs, located on either side of the vertebral column in the abdominal cavity. Each kidney is about 10–12 cm long and highly vascular, receiving blood from the renal arteries.

- **Function:**
 - Filter blood to remove metabolic wastes such as urea, uric acid, and creatinine.
 - Regulate water balance, electrolytes (Na^+ , K^+ , Cl^-), and pH.
 - Produce hormones such as **erythropoietin** (stimulates red blood cell production) and **renin** (regulates blood pressure).
- Damage to kidneys (renal failure) may require dialysis, an artificial method of filtering the blood.



2. Ureters – Urine Transport Tubes

The ureters are two narrow muscular tubes (25–30 cm long) that carry urine from each kidney to the urinary bladder. They use **peristaltic contractions** (wave-like muscular movements) to ensure continuous flow of urine, regardless of body position.

- **Function:** Safely transport urine without backflow.
- **Example:** Blockage of ureters by kidney stones can cause severe pain (renal colic) and obstruction of urine flow.

3. Urinary Bladder – Temporary Storage Organ

The urinary bladder is a hollow, muscular sac located in the pelvic cavity. It temporarily stores urine until it is excreted. The inner wall of the bladder is lined with a special tissue called **transitional epithelium**, which allows stretching as the bladder fills.

- **Capacity:** In adults, the bladder can hold about **400–600 mL** of urine.
- **Function:** Provides voluntary control of urination (micturition) through relaxation of sphincter muscles. Urinary tract infections (UTIs) often begin in the bladder (cystitis), especially in females due to their shorter urethra.

4. Urethra – The Exit Passage

The urethra is a narrow tube that carries urine from the urinary bladder to the outside of the body.

- **In males:** It is about 20 cm long and also serves as a passage for semen during reproduction.
- **In females:** It is much shorter (about 4 cm), which makes females more prone to urinary infections.
- **Function:** Provides the final passageway for urine elimination.

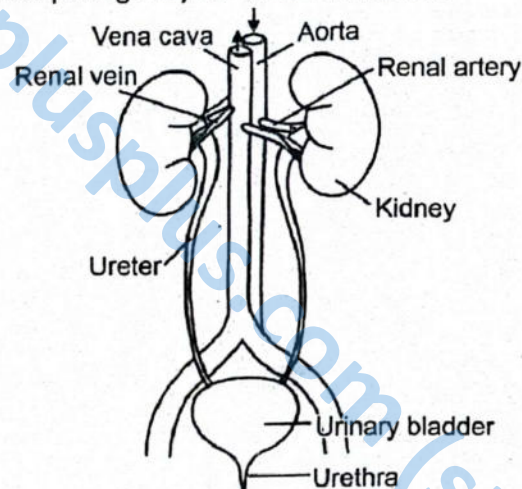


Figure: Excretory system of man

SLO [B-12-R-12]

Describe the structure of the kidney.

STRUCTURE OF THE KIDNEY

The **kidneys** are the primary excretory organs of humans, responsible for filtering blood, maintaining water and electrolyte balance, and excreting nitrogenous wastes. Each kidney is a highly specialized organ with a complex structure that supports these vital functions. Understanding the structure of the kidney is crucial for appreciating how it performs its role in excretion and homeostasis.

1. External Structure

Shape, Size, and Location:

Each kidney is **bean-shaped**, reddish-brown in color, and measures about **10–12 cm in length**, **5–7 cm in width**, and **3–4 cm in thickness** in adults. They weigh approximately **120–150 grams** each. The kidneys are located in the **posterior abdominal cavity** on either side of the vertebral column, just below the ribcage. The right kidney is usually slightly lower than the left due to the presence of the liver.



Hilum and Renal Capsule:

On the inner (medial) surface of each kidney is a vertical depression called the **hilum**. Through the hilum, the **renal artery enters**, and the **renal vein and ureter exit**. The entire kidney is enclosed in a **tough fibrous capsule** called the **renal capsule**, which provides protection against physical trauma and infections. Surrounding this capsule is a layer of **adipose (fatty) tissue**, known as the **perirenal fat**, which cushions and holds the kidney in place.

2. Internal Structure

When cut longitudinally, the kidney reveals three distinct regions: **renal cortex, renal medulla, and renal pelvis**.

(a) Renal Cortex:

The outermost region is called the **renal cortex**, which appears granular due to the presence of numerous **renal corpuscles and convoluted tubules** of nephrons. This is the site where blood filtration begins.

(b) Renal Medulla:

Beneath the cortex lies the **renal medulla**, a darker, striated region arranged into **renal pyramids**. Each pyramid is cone-shaped, with its broad base facing the cortex and its tip (called **renal papilla**) projecting into minor calyces. The striated appearance is due to the presence of **loops of Henle and collecting ducts** running parallel.

(c) Renal Pelvis:

At the innermost part of the kidney lies a funnel-shaped chamber called the **renal pelvis**. It collects urine from the calyces and narrows down to form the **ureter**, which carries urine to the bladder.

- **Calyces (minor and major):** The renal papillae drain urine into small cup-like cavities called **minor calyces**. These join to form **major calyces**, which then open into the renal pelvis.

3. Nephrons – The Functional Units

The **nephron** is the microscopic structural and functional unit of the kidney. Each kidney contains about **1–1.5 million nephrons**, collectively responsible for urine formation.

Components of a Nephron:

- **Renal Corpuscle:** Consists of the **Bowman's capsule** (a double-walled cup-like structure) surrounding a tuft of capillaries known as the **glomerulus**. This is the site of blood filtration.
- **Renal Tubule:** After the Bowman's capsule, the filtrate passes through:
 - **Proximal convoluted tubule (PCT):** Responsible for reabsorption of water, glucose, amino acids, and ions.
 - **Loop of Henle:** A U-shaped structure dipping into the medulla; crucial for concentrating urine.
 - **Distal convoluted tubule (DCT):** Involved in selective reabsorption and secretion.
 - **Collecting duct:** Receives urine from many nephrons and drains into renal pelvis.

4. Blood Supply of the Kidney

The kidney is richly supplied with blood, receiving about **20–25% of cardiac output**. Blood enters through the **renal artery**, which branches into smaller arterioles leading to the glomerulus. After filtration, blood exits via the **renal vein**. The unique arrangement of two capillary beds in series (glomerular and peritubular capillaries) ensures efficient filtration and reabsorption.

5. Protective and Supportive Features

The kidneys are protected by multiple layers:

1. **Renal capsule** – tough fibrous layer directly covering the kidney.
2. **Perirenal fat** – cushions the organ.
3. **Renal fascia** – connective tissue anchoring the kidneys to surrounding structures.

These ensure the kidney remains securely in place and safe from mechanical shocks.

The kidney is a **highly organized organ** consisting of an external cortex, an inner medulla with pyramids, and a central pelvis that collects urine. Its microscopic units, the **nephrons**, are specialized for filtration, reabsorption, and secretion, ensuring waste removal and homeostasis. The complex structure of the kidney directly relates to its essential role in maintaining the internal environment of the body.



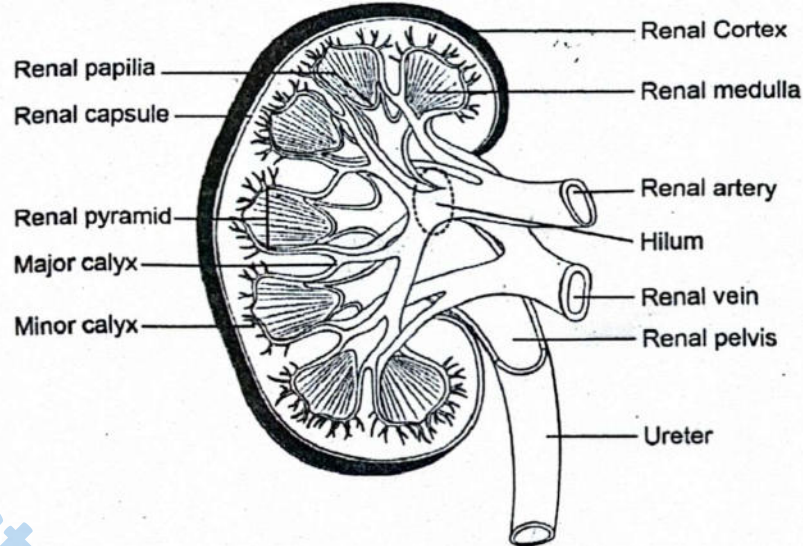


Figure: Internal anatomy of the kidney

SLO [B-12-R-13]

Relate the structure of kidney with its function.

RELATIONSHIP BETWEEN STRUCTURE AND FUNCTION OF KIDNEY

The human kidney is a structurally complex organ designed for the precise filtration of blood and maintenance of internal chemical balance. Each anatomical component of the kidney plays a specific role in the processes of filtration, reabsorption, secretion, and excretion. The major anatomical parts of the kidney and their corresponding functions are outlined below:

1. Renal Capsule

The **renal capsule** is the outermost protective covering of the kidney. It comprises **three layers**—the **renal fascia**, **perirenal fat**, and the **fibrous capsule**—which collectively:

- Shield the kidney from mechanical injury and infections.
- Anchor the kidney in place within the abdominal cavity.
- Provide structural support and insulation.

2. Renal Artery

The **renal artery** is a major blood vessel that delivers oxygenated, unfiltered blood from the abdominal aorta to the kidneys. It:

- Branches extensively within the kidney to supply blood to each nephron.
- Delivers approximately **1.2 liters of blood per minute** to the kidneys at rest, allowing continuous filtration of plasma and waste removal.

3. Renal Cortex

The **renal cortex** forms the **outer region** of the kidney and is richly vascularized. It serves as the **primary site** for:

- **Filtration of blood** via structures known as **nephrons**, specifically the Bowman's capsules and glomeruli.
- **Production of erythropoietin (EPO)**, a hormone that stimulates red blood cell formation in the bone marrow in response to low oxygen levels.

4. Renal Medulla

Located deep to the cortex, the **renal medulla** contains cone-shaped structures known as **renal pyramids**. It functions in:

- **Concentration and collection of urine** through nephron segments such as the loops of Henle and collecting ducts.
- **Osmoregulation**, through the countercurrent mechanism that concentrates urine and conserves water.



5. Renal Papilla

The **renal papilla** is the tapered apex of each renal pyramid. It:

- Serves as the point where urine produced in the medulla is **drained into the minor calyces**.
- Acts as a functional interface between the medullary collecting ducts and the renal pelvis.

6. Renal Pelvis

The **renal pelvis** is a **funnel-shaped cavity** situated in the renal hilum. It:

- Collects urine from the major calyces and channels it into the **ureter**.
- Functions as the final receiving chamber before urine leaves the kidney.

7. Renal Vein

The **renal vein** carries filtered, deoxygenated blood **away from the kidney** and returns it to the **inferior vena cava**. Each kidney has its own renal vein. This vessel:

- Completes the renal circulation loop after blood has been processed by nephrons.
- Ensures **metabolically balanced blood** returns to systemic circulation.

This structural organization enables the kidneys to efficiently perform their roles in **filtration, homeostasis, blood pressure regulation, and hormone secretion**. Each component contributes to the integrated function of this vital organ system.

SLO [B-12-R-14]

Explain the detailed structure of a nephron.

STRUCTURE OF A NEPHRON

The **nephron** is the fundamental structural and functional unit of the kidney, responsible for the filtration of blood and the formation of urine. Each human kidney contains approximately **1 to 1.5 million nephrons**, which operate collectively to maintain fluid and electrolyte balance, remove waste products, and regulate blood pH.

Structure of the Nephron

A typical nephron consists of two main parts:

- Renal corpuscle
- Renal tubule

1. Renal Corpuscle

The **renal corpuscle** is located in the renal cortex and is the site of initial blood filtration. It comprises:

- **Glomerulus** – A dense, tuft-like network of capillaries that receives blood from the **afferent arteriole**.
- **Bowman's capsule** – A double-walled, cup-like structure that encases the glomerulus. It marks the **beginning** of the renal tubule and collects the filtrate from the glomerular capillaries.

2. Renal Tubule and Associated Structures

After the filtrate is collected in Bowman's capsule, it passes through a series of specialized tubules where selective reabsorption and secretion occur:

a. Proximal Convoluted Tubule (PCT)

Immediately following Bowman's capsule, the filtrate enters the **proximal convoluted tubule**, a **highly coiled segment** located in the cortex. This region is responsible for the **reabsorption of water, ions, glucose, and amino acids** into the surrounding capillaries.

b. Loop of Henle

The PCT descends into the renal medulla to form the **loop of Henle**, which has two distinct limbs:

- **Descending limb** – Permeable to water, facilitating water reabsorption.
- **Ascending limb** – Impermeable to water but actively transports salts out of the filtrate.

*This arrangement enables the kidney to **concentrate urine** through the countercurrent multiplier mechanism.*

c. Distal Convoluted Tubule (DCT)

As the loop of Henle ascends back toward the cortex, it transitions into the **distal convoluted tubule**, another coiled segment. The DCT fine-tunes ion concentration and contributes to **acid-base balance** and **secretion of certain wastes**.



d. **Collecting Duct System**

The DCT opens into a **collecting duct**, which receives filtrate from multiple nephrons.

- Several collecting ducts merge to form larger ducts that drain into the **renal pelvis** via the **renal papillae**.
- The collecting system plays a key role in **water reabsorption** under **hormonal control** (e.g., antidiuretic hormone).

Blood Circulation through the Nephron

An efficient and specialized blood supply supports nephron function, ensuring continuous filtration and reabsorption.

- **Renal artery** branches into **afferent arterioles**, each supplying a glomerulus within a renal corpuscle.
- **Efferent arterioles**, which emerge from the glomerulus, do not carry blood directly to veins. Instead, they form a **peritubular capillary network** around the proximal and distal tubules in the cortex.
- In nephrons with long loops of Henle (juxtamedullary nephrons), specialized capillaries called the **vasa recta** descend into the medulla, running parallel to the loops of Henle. This arrangement aids in the maintenance of the medullary osmotic gradient.
- The peritubular capillaries and vasa recta eventually drain into **venules**, which converge into the **renal vein**, returning filtered blood to the **inferior vena cava**.

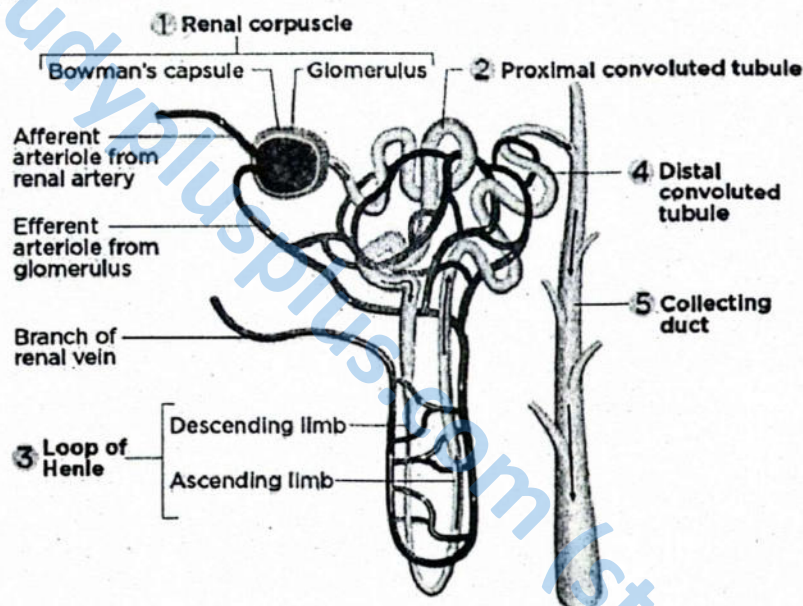


Figure: Structure of nephron

SLOs Based (MCQs)

STRUCTURE OF KIDNEY AND NEPHRON

- | | |
|---|---|
| <p>1. Which of the following is the functional unit of the kidney?</p> <p>A. Glomerulus B. Nephron
C. Bowman's capsule D. Loop of Henle</p> <p>2. What is the main function of Bowman's capsule?</p> <p>A. Reabsorption of glucose B. Secretion of waste
C. Filtration of blood plasma D. Transport of urine</p> <p>3. In which region of the nephron does the majority of selective reabsorption occur?</p> <p>A. Collecting duct B. Loop of Henle
C. Proximal convoluted tubule
D. Distal convoluted tubule</p> | <p>4. Which blood vessel carries blood out of the glomerulus?</p> <p>A. Afferent arteriole B. Renal artery
C. Renal vein D. Efferent arteriole</p> <p>5. The loop of Henle primarily functions in:</p> <p>A. Filtering blood B. Secreting waste
C. Producing hormones D. Concentrating urine</p> <p>6. What is the role of the renal pelvis?</p> <p>A. Filters nitrogenous waste
B. Stores urine
C. Collects urine and funnels it into the ureter
D. Produces erythropoietin</p> |
|---|---|



7. The vasa recta is functionally associated with:
 A. Bowman's capsule B. Renal artery
 C. Proximal tubule D. Loop of Henle
8. Which part of the nephron is directly connected to the collecting duct?
 A. Proximal convoluted tubule
 B. Loop of Henle C. Glomerulus
 D. Distal convoluted tubule

9. Which kidney structure encases and protects the kidney?
 A. Renal pelvis B. Renal capsule
 C. Renal medulla D. Renal cortex
10. Blood from the kidney eventually drains into:
 A. Hepatic vein B. Pulmonary vein
 C. Inferior vena cava D. Superior vena cava

Answer Key

1. B 2. C 3. C 4. D 5. D 6. C 7. D 8. D 9. B 10. C

Short Questions

Q.1 What is a nephron, and what are its main parts?

Ans. A nephron is the functional unit of the kidney, responsible for blood filtration and urine formation. Each nephron has two main components:

1. **Renal corpuscle** – made up of the glomerulus and Bowman's capsule, where filtration occurs.
2. **Renal tubule** – consisting of the proximal convoluted tubule (PCT), loop of Henle, distal convoluted tubule (DCT), and the collecting duct, where reabsorption and secretion take place.

MARKING SCHEME (3 Marks)

- Definition (1), parts named (2).

Q.2 What happens in the proximal convoluted tubule (PCT)?

Ans. The PCT is the first part of the renal tubule, located after Bowman's capsule. It is the site of **extensive reabsorption**, where essential substances such as glucose, amino acids, ions (Na^+ , Cl^- , HCO_3^-), and a significant amount of water are reabsorbed into the surrounding capillaries. This process ensures that useful materials are returned to the bloodstream.

MARKING SCHEME (3 Marks)

- Location (1), functions (2).

Q.3 Describe the loop of Henle and its function.

Ans. The **loop of Henle** is a U-shaped section of the nephron tubule. It has two limbs:

- **Descending limb** – permeable to water but not salts; water diffuses out.
- **Ascending limb** – impermeable to water but actively transports salts into the medulla.

Together, these create a **concentration gradient in the medulla**, which is crucial for **water conservation** and producing concentrated urine.

MARKING SCHEME (3 Marks)

- Location in nephron (1), functions (2).

Q.4 What is the role of the distal convoluted tubule (DCT)?

Ans. The DCT, located after the loop of Henle, fine-tunes urine composition. It reabsorbs ions such as Na^+ and Ca^{2+} , helps maintain **acid-base balance** by secreting H^+ or reabsorbing HCO_3^- , and contributes to selective secretion of wastes. It also responds to **hormones** such as aldosterone, which regulates sodium reabsorption.

MARKING SCHEME (3 Marks)

- Location (1), functions (2).

Q.5 Explain the structure and function of the collecting duct.

Ans. The **collecting duct** is a long tubular structure that receives urine from multiple nephrons and carries it through the renal medulla to the renal pelvis. Its walls are responsive to **antidiuretic hormone (ADH)**, which increases water reabsorption. This allows the body to regulate **urine concentration and volume**, especially during dehydration.

MARKING SCHEME (3 Marks)

- Structure (1), functions (2).

Q.6 How is blood supplied to and from the nephron?

Ans. Blood enters the nephron through the **afferent arteriole**, which delivers blood to the glomerulus for filtration. The filtered blood leaves via the **efferent arteriole**, which branches into the **peritubular capillaries** and the **vasa recta**. These capillaries surround the nephron, allowing reabsorption and secretion, before draining into venules and eventually the renal vein.

MARKING SCHEME (3 Marks)

- Role of efferent & afferent arteriole (1), pathway (2).



Q.7 What is the role of Bowman's capsule?

Ans. The **Bowman's capsule** is a double-walled cup-like structure located in the renal cortex. It surrounds the glomerulus and serves as the **initial site of urine formation**. Its main function is to **collect the filtrate** produced during glomerular filtration and direct it into the renal tubule for further modification.

MARKING SCHEME (3 Marks)
• Location (1), function (2).

Q.8 What is the vasa recta and its function?

Ans. The **vasa recta** is a specialized network of capillaries that runs parallel to the loop of Henle in **juxtamedullary nephrons**. Its primary role is to **maintain the osmotic gradient** in the medulla through countercurrent exchange, which is vital for **concentrated urine formation** and water conservation.

MARKING SCHEME (3 Marks)
• Structure (1), function (2).

Q.9 Describe the renal corpuscle and its function.

Ans. The **renal corpuscle**, located in the renal cortex, consists of two parts: the **glomerulus** (a tuft of capillaries) and **Bowman's capsule** (the surrounding cup-like structure). Its main function is **filtration of blood plasma**, producing the initial filtrate that enters the nephron for further processing into urine.

MARKING SCHEME (3 Marks)
• Location (1), function (2).

Q.10 What role does the renal pelvis play in the urinary system?

Ans. The **renal pelvis** is a funnel-shaped cavity located at the center of the kidney. It collects urine from the **major calyces** and channels it into the **ureter**. Thus, it acts as the **final collecting site** within the kidney before urine passes to the bladder for storage.

MARKING SCHEME (3 Marks)
• Location (1), function (2).

SLO [B-12-R-15]

Explain the process of glomerular filtration, selective re-absorption and tubular secretion as the events in kidney functioning.

URINE FORMATION

Urine formation in the nephron is a highly coordinated process that ensures the removal of metabolic wastes and the regulation of blood composition. It involves three main stages:

1. Glomerular Filtration
2. Tubular Reabsorption
3. Tubular Secretion

1. Glomerular Filtration (Ultrafiltration under Pressure)

Location: Renal corpuscle (Glomerulus + Bowman's capsule)

Mechanism: Passive filtration under pressure

Glomerular filtration is the **first step** in urine formation. It occurs in the **Bowman's capsule**, where blood from the **afferent arteriole** enters a bundle of capillaries known as the **glomerulus**.

► Hydrostatic Pressure Drives Filtration

Filtration occurs due to **high hydrostatic pressure**, which is generated by the incoming blood. This pressure is unusually high in the glomerulus—**significantly higher than in other capillary beds** of the body.

This is achieved because the **efferent arteriole** (outgoing) is **narrower than the afferent arteriole** (incoming). As blood is forced into a narrower vessel, **pressure builds up**, pushing plasma and small solutes out of the capillaries and into the Bowman's capsule.

Substances which are filtered:

Filtered into the capsule:

- Water
- Ions (e.g., Na^+ , Cl^- , K^+)
- Glucose
- Amino acids
- Urea and small nitrogenous waste molecules



Remain in the blood (not filtered):

- Proteins (too large)
- Red blood cells and platelets (cellular components)

The resulting fluid in the Bowman's capsule is called **glomerular filtrate**. It resembles **blood plasma** in composition, except that it lacks proteins and blood cells.

2. Tubular Reabsorption (Selective Reabsorption)

Location: Mainly proximal convoluted tubule (PCT); also in loop of Henle, DCT, and collecting duct

Mechanism: Active and passive transport

After filtration, the body must recover useful substances that were accidentally filtered out. This is accomplished through **selective reabsorption**—a process by which essential solutes are **actively or passively transported** back into the blood via the surrounding capillaries.

► Substances Reabsorbed Include:

- 100% of glucose and most amino acids
- Vitamins
- Inorganic salts (e.g., Na^+ , Cl^- , HCO_3^-)
- Variable amounts of water, depending on hydration status

The **proximal convoluted tubule** is the major site of reabsorption. The loop of Henle and distal convoluted tubule further fine-tune the absorption of salts and water.

3. Tubular Secretion (pH Regulation and Waste Elimination)

Location: Primarily in the distal tubule and loop of Henle

Mechanism: Active secretion by epithelial cells

Tubular secretion is the **final step** in urine formation. It involves the **active transport of certain substances** from the peritubular capillaries directly into the renal tubule. These substances were not removed during filtration or were produced in the tubule cells themselves.

► Substances Secreted:

- **Hydrogen ions (H^+)** — help regulate blood and urine pH
- **Ammonium ions (NH_4^+)** — detoxify nitrogenous waste
- **Potassium ions (K^+)** and certain drugs/toxins

This process is essential for:

- Acid-base balance
- Fine-tuning of electrolyte levels
- Excretion of substances in excess

The normal urine pH ranges from **4.8 to 7.5**, reflecting the body's dynamic response to acid-base fluctuations.

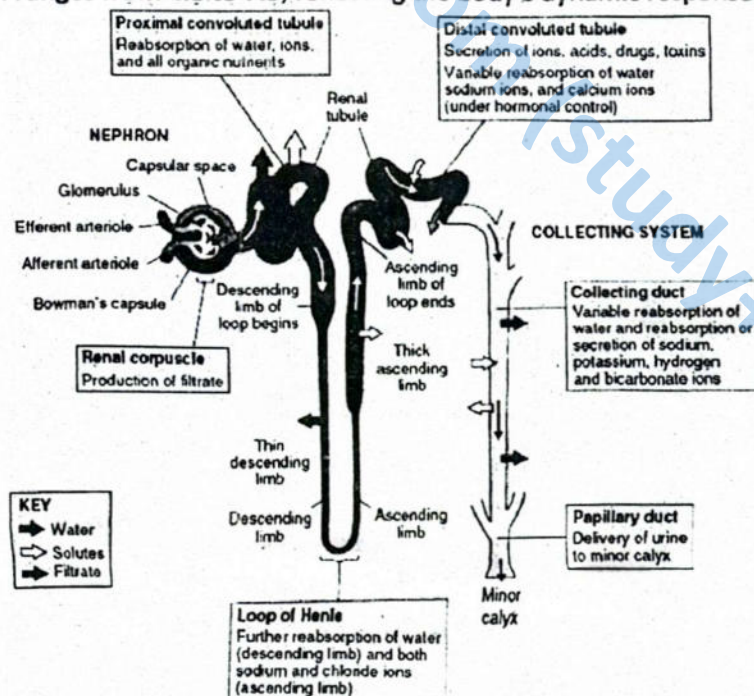


Figure: Urine formation

There are two types of nephrons, cortical nephrons and juxtamedullary nephrons. Cortical nephrons are found in the cortex. They have their renal corpuscle in the superficial renal cortex and have relatively short loops of Henle. 70 to 80% nephrons in human kidney are cortical. Under normal conditions of water availability, the cortical nephrons deal with the control of blood volume by forming diluted urine. Juxtamedullary nephrons have their renal corpuscle close to the junction of the cortex and medulla. They have long loop of Henle which extends deep into the medulla. These types of nephrons are relatively rare and only comprise 20-30% of the nephrons in the human kidney. When water is in short supply, increased water retention occurs through juxtamedullary nephrons.

SLO [B-12-R-16]

Explain regulatory mechanism involved in concentration of urine.

MECHANISM OF URINE CONCENTRATION

Mechanism of Urine Concentration: The Countercurrent Multiplier System

The kidneys must conserve water when the body is dehydrated and remove excess water when it is overhydrated. The ability to produce concentrated (**hypertonic**) urine is essential for water balance and survival in terrestrial environments.

This is made possible by a specialized mechanism called the **countercurrent multiplier**, which operates in the **loop of Henle** and is supported by the **vasa recta**—the capillary network surrounding the nephron.

Role of the Loop of Henle and the Vasa Recta

The **loop of Henle** plays a central role in creating a gradient of solute concentration in the kidney medulla. The descending and ascending limbs of the loop differ in their permeability:

- **Descending limb:** Permeable to water, but not to solutes.
- **Ascending limb:** Impermeable to water but actively transports **sodium chloride (NaCl)** into the surrounding tissue.

This arrangement establishes a **countercurrent system**, where filtrate flows in opposite directions in the two limbs of the loop. Simultaneously, **blood in the vasa recta** flows counter to the flow in the nephron, helping maintain the osmotic gradient.

Steps of the Countercurrent Multiplier Mechanism

1. Active NaCl Transport:

Sodium chloride is **actively pumped out** of the ascending limb into the medullary interstitium. This increases the solute concentration in the surrounding tissue, making it **hypertonic**.

2. Water Reabsorption by Osmosis:

As filtrate flows down the descending limb, **water moves out** into the hypertonic interstitial fluid by **osmosis**, concentrating the filtrate.

3. Flow Continuation and Exchange:

This continuous flow and differential permeability create a **vertical osmotic gradient** in the medulla. As a result, **about 99.5% of water** is reabsorbed before urine reaches the ureter.

4. Role of Hormones:

- **Aldosterone:** Regulates sodium reabsorption, especially in the distal tubule and collecting duct.
- **Antidiuretic Hormone (ADH):** Secreted from the **posterior pituitary**, ADH increases the **permeability of the collecting duct to water**, allowing more water to be reabsorbed into the bloodstream.

SCIENCE TITBITS

In addition to their excretory and osmoregulatory role, kidneys also help to control the red blood cell formation by secreting the hormone erythropoietin and help to regulate blood pressure by secreting the (EPO) enzyme renin.

Formation of Hypertonic Urine

In dehydration or water deficiency, **high ADH levels** promote maximum water reabsorption in the collecting duct, producing a **small volume of concentrated urine**. This is critical for conserving body fluids.



In contrast, during overhydration, ADH secretion is suppressed, leading to less water reabsorption and the production of dilute urine.

Summary of Process of Urine Concentration

Component	Role
Loop of Henle	Establishes osmotic gradient via countercurrent flow
Descending limb	Reabsorbs water via osmosis
Ascending limb	Pumps out NaCl; impermeable to water.
Vasa recta	Maintains medullary gradient via countercurrent exchange
ADH (Antidiuretic Hormone)	Increases water reabsorption in collecting duct
Aldosterone	Enhances Na ⁺ reabsorption, indirectly aiding water retention.

SLO [B-12-R-17]

Justify the functioning of kidneys as both excretion and osmoregulation.

KIDNEYS AS ORGANS OF EXCRETION AND OSMOREGULATION

The kidneys are vital organs of the urinary system that play a dual role in maintaining the body's internal balance. They function not only as **excretory organs**, removing waste products of metabolism, but also as **osmoregulatory organs**, regulating the concentration of water and salts in body fluids. These two roles are closely linked, as efficient excretion requires controlled regulation of fluid and electrolyte balance.

Kidneys as Excretory Organs

Excretion is the process of removing metabolic wastes from the body. The kidneys filter blood through specialized structures called **nephrons**, which separate useful substances from harmful wastes.

1. Removal of Nitrogenous Wastes:

- Proteins and nucleic acids are metabolized to produce nitrogenous wastes such as **urea, creatinine, and uric acid**.
- These compounds are toxic if allowed to accumulate, so the kidneys filter them out and excrete them in urine.

2. Detoxification of the Blood:

- Drugs, toxins, and by-products of metabolism (e.g., ammonia converted to urea in the liver) are eliminated.
- This purification ensures that blood leaving the kidneys is free from harmful metabolic end products.

3. Acid-Base Balance:

- By excreting hydrogen ions (H⁺) and reabsorbing bicarbonate (HCO₃⁻), kidneys maintain the blood pH within a narrow range (around 7.35–7.45).
- Without this, metabolic acidosis or alkalosis could impair cellular functions.

Thus, the kidneys serve as highly efficient filters, protecting the body from toxic buildup and maintaining chemical stability.

Kidneys as Osmoregulatory Organs

Osmoregulation refers to the control of water and salt balance in the body to ensure homeostasis. Kidneys maintain the **osmotic pressure of blood plasma** and regulate extracellular fluid composition.

1. Regulation of Water Balance:

- The **antidiuretic hormone (ADH)** increases the permeability of the collecting ducts, allowing more water to be reabsorbed into the bloodstream during dehydration.
- In conditions of excess water intake, less ADH is secreted, resulting in dilute urine.

2. Regulation of Salt (Electrolyte) Balance:

- The kidneys control the levels of sodium, potassium, chloride, and calcium ions.
- The hormone **aldosterone**, secreted by the adrenal cortex, promotes sodium reabsorption and potassium excretion in the distal tubule, which also indirectly regulates water retention.



3. Maintenance of Blood Pressure and Volume:

- Specialized kidney cells release **renin**, which triggers the renin–angiotensin–aldosterone system (RAAS). This pathway increases blood pressure by vasoconstriction and stimulating sodium and water reabsorption.
- In this way, kidneys play a central role in long-term regulation of blood pressure.

► Integration of Excretion and Osmoregulation

The excretory and osmoregulatory functions of kidneys are interdependent. For example:

- When waste products are excreted, water is also removed. The kidneys must regulate the amount of water lost to prevent dehydration.
- Similarly, regulation of sodium ions not only maintains osmotic balance but also determines the volume of urine produced and the concentration of wastes.

This dual role allows the kidneys to maintain **homeostasis** — a stable internal environment despite external changes. For instance, during intense exercise, increased metabolic wastes (urea, lactic acid) must be excreted, while at the same time water and electrolytes lost in sweat must be balanced by kidney regulation.

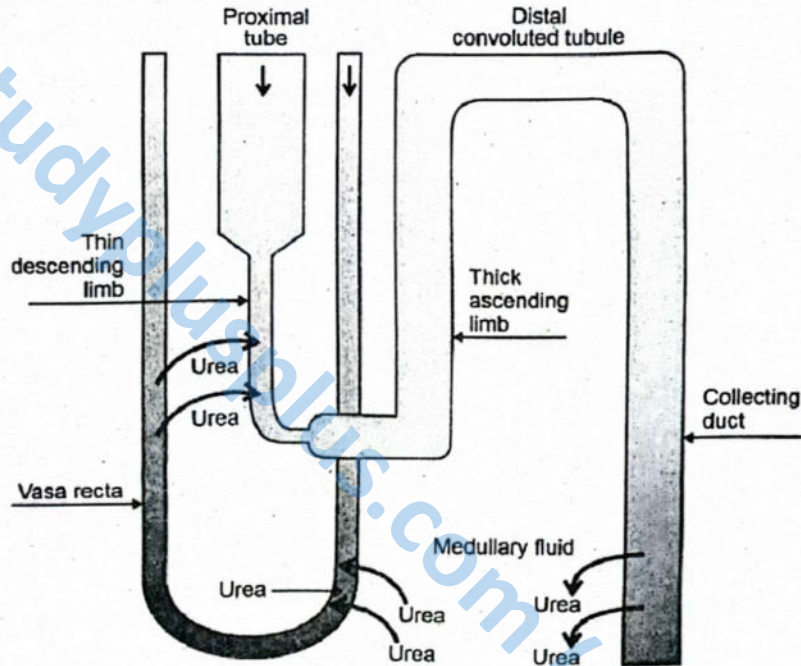


Figure: Counter current mechanism

SLO [B-12-R-18]

Compare the function of two major capillary beds in kidneys i.e., glomerular capillaries and peritubular capillaries.

CAPILLARY BEDS OF THE KIDNEY

The kidney is unique in having **two successive capillary networks** associated with each nephron:

- **Glomerular capillaries** – specialized for filtration.
- **Peritubular capillaries (including vasa recta in juxtamedullary nephrons)** – specialized for reabsorption and exchange.

These capillaries are connected by **efferent arterioles**, unlike most organs where arterioles connect capillaries to venules. This arrangement ensures both efficient filtration and recovery of useful substances.

1. Glomerular Capillaries

Structure

- Located inside the **Bowman's capsule** of the nephron.

- Composed of **fenestrated capillaries**, supported by a basement membrane and podocytes.
- Supplied by the **afferent arteriole** (wider) and drained by the **efferent arteriole** (narrower), maintaining high pressure.

Function

- Specialized for ultrafiltration of blood.
- High hydrostatic pressure (~55 mmHg) forces water, electrolytes, glucose, amino acids, and urea out of blood into Bowman's capsule.
- Large molecules such as proteins and blood cells remain in the capillaries.

Glomerular capillaries act like a biological sieve, forming **glomerular filtrate** that resembles blood plasma but is protein-free.

2. Peritubular Capillaries

Structure

- Network of capillaries surrounding the proximal tubule, loop of Henle, distal tubule, and collecting duct in the renal cortex.
- Arise from the efferent arteriole of the glomerulus.
- Pressure here is low, favouring reabsorption.
- In juxtamedullary nephrons, specialized long capillaries called the vasa recta run parallel to the loop of Henle into the medulla.

Function

- Major role is reabsorption and secretion.
- Reabsorb nearly 99% of water and useful solutes (like glucose, amino acids, sodium, and bicarbonate) from filtrate back into the blood.
- Participate in secretion of waste products (e.g., hydrogen ions, potassium, drugs) into the tubules.
- Vasa recta maintain the osmotic gradient of the renal medulla, crucial for urine concentration.

Peritubular capillaries act as recovery channels, ensuring that essential substances are reclaimed and only wastes are excreted.

Comparison of Functions

Feature	Glomerular Capillaries	Peritubular Capillaries
Location	Inside Bowman's capsule	Around renal tubules (cortex and medulla)
Pressure	High (≈ 55 mmHg)	Low ($\approx 10-15$ mmHg)
Specialization	Filtration	Reabsorption & secretion
End Product	Glomerular filtrate	Recovery of water, nutrients, ions; final urine composition
Unique Adaptation	Fenestrated structure with podocytes	Vasa recta help maintain osmotic gradient in medulla

SLOs Based (MCQs)

URINE FORMATION & COUNTER CURRENT MULTIPLIER MECHANISM

- | | |
|--|---|
| <p>1. What is the primary force responsible for glomerular filtration in the renal capsule?</p> <p>A. Osmotic pressure
B. Atmospheric pressure
C. Hydrostatic pressure
D. Renal tubular pressure</p> <p>2. Which of the following substances is <i>not</i> normally filtered into the glomerular filtrate?</p> <p>A. Glucose
B. Sodium ions
C. Plasma proteins
D. Urea</p> <p>3. Why is the blood pressure in glomerular capillaries unusually high?</p> | <p>A. Blood enters from renal vein
B. Afferent arteriole is narrower than efferent arteriole
C. Efferent arteriole is narrower than afferent arteriole
D. Presence of valves in renal capsule</p> <p>4. Which part of the nephron is primarily responsible for selective reabsorption?</p> <p>A. Glomerulus
B. Proximal convoluted tubule
C. Descending limb of loop of Henle
D. Renal pelvis</p> |
|--|---|



5. Which hormone increases the permeability of the collecting duct to water?
A. Aldosterone
B. ADH (antidiuretic hormone)
C. Renin
D. Insulin
6. What is the main function of tubular secretion?
A. To remove glucose from the filtrate
B. To balance oxygen concentration
C. To eliminate substances like hydrogen and ammonium ions
D. To absorb proteins back into the blood
7. Which statement best explains the term "countercurrent" in the kidney?
A. Movement of water against gravity
B. Flow of filtrate and blood in the same direction
C. Opposite flow of filtrate in nephron and blood in vasa recta
8. In which part of the nephron is sodium chloride actively transported out into the medullary interstitium?
A. Descending limb of loop of Henle
B. Ascending limb of loop of Henle
C. Proximal convoluted tubule
D. Renal capsule
9. Which of the following contributes to the formation of hypertonic urine?
A. Increased blood glucose
B. Reduced filtration pressure
C. Increased ADH secretion
D. Decreased Na^+ reabsorption
10. What is the pH range of normal urine?
A. 6.8 – 9.5
B. 4.8 – 7.5
C. 3.0 – 4.5
D. 7.0 – 9.0
11. Which of the following correctly describes the pressure filtration mechanism in the renal corpuscle?

Option	Mechanism
A	Filtration occurs due to osmotic pressure generated by glomerular capillaries.
B	Filtration occurs because efferent arteriole has a narrower diameter than afferent arteriole.
C	Filtration occurs due to active transport of solutes into the Bowman's capsule.
D	Filtration is caused by secretion of ADH from the pituitary gland.

12. Which of the following options best explains the countercurrent multiplier mechanism in the loop of Henle?

Option	Mechanism
A	Sodium chloride is reabsorbed passively in descending limb, increasing medullary osmolarity.
B	Water is actively reabsorbed from ascending limb, forming dilute urine.
C	Sodium chloride is actively transported from ascending limb, increasing interstitial solute concentration.
D	Water enters ascending limb by osmosis, creating concentrated filtrate.

13. Which process contributes most to the formation of hypertonic urine during dehydration?

Option	Description
A	Reabsorption of glucose and amino acids from distal tubule.
B	Secretion of hydrogen ions into the glomerulus.
C	ADH-induced water reabsorption from collecting duct due to high medullary osmolarity.
D	Loss of sodium from filtrate into Bowman's capsule.

14. Which table best describes the direction of flow in the countercurrent exchange system of the kidney?

Option	Filtrate in Loop of Henle	Blood in Vasa Recta
A	Descending → Ascending	Ascending → Descending
B	Descending → Ascending	Descending → Ascending
C	Ascending → Descending	Ascending → Descending
D	Ascending → Descending	Descending → Ascending

15. Which of the following mechanisms explains selective reabsorption in the nephron?

Option	Description
A	All substances are reabsorbed by simple diffusion.
B	Water and urea are reabsorbed by active transport in the loop of Henle.
C	Useful substances like glucose, amino acids, and salts are reabsorbed by active and passive means.
D	Only waste products are reabsorbed to maintain osmotic balance.

Answer Key

1. C	2. C	3. C	4. B	5. B	6. C	7. C	8. B	9. C	10. B	11. B	12. C	13. C	14. A	15. C
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Q.1 Define glomerular filtration. Where does it occur?

Ans. Glomerular filtration is the process in which **blood plasma is filtered under pressure** across the glomerular capillaries into Bowman's capsule. It is driven by **hydrostatic pressure**, which forces water and small solutes (like glucose, amino acids, and ions) into the filtrate while leaving larger molecules behind. This process takes place in the **renal corpuscle (Bowman's capsule)**.

MARKING SCHEME (3 Marks)
• Definition (2), location (1).

Q.2 Why are proteins and blood cells not found in glomerular filtrate?

Ans. Proteins and blood cells are **too large to pass through the filtration membrane** formed by the glomerular capillary walls, basement membrane, and podocyte slits. Therefore, they remain in the bloodstream. This selective filtration ensures that only small, useful molecules enter the filtrate while essential blood components are retained.

MARKING SCHEME (3 Marks)
• Importance of process (1), reason (2).

Q.3 What is selective reabsorption and where does it primarily take place?

Ans. Selective reabsorption is the **process of actively and passively reclaiming essential substances** such as glucose, amino acids, vitamins, salts, and most water from the filtrate back into the blood. This prevents the loss of valuable nutrients. It primarily occurs in the **proximal convoluted tubule (PCT)** of the nephron.

MARKING SCHEME (3 Marks)
• Definition (2), site (1).

Q.4 What is the function of tubular secretion?

Ans. Tubular secretion is the **active transfer of specific substances** (e.g., hydrogen ions, potassium ions, ammonium ions, drugs, and toxins) from the blood into the tubular fluid. Its main functions are to **maintain blood pH**, remove waste products not filtered earlier, and aid in regulating electrolyte balance.

MARKING SCHEME (3 Marks)
• Definition (1), functions (2).

Q.5 Describe the role of the loop of Henle in urine concentration.

Ans. The loop of Henle establishes an osmotic gradient in the renal medulla through the countercurrent multiplier mechanism. In this process:

- The **descending limb** is permeable to water, allowing it to move out by osmosis.
- The **ascending limb** actively transports sodium and chloride ions into the interstitial fluid but is impermeable to water.
- This gradient enables the kidney to produce **concentrated urine** and conserve water.

MARKING SCHEME (3 Marks)
• Location (1), mechanism explained (2).

Q.6 What is the function of ADH in the nephron?

Ans. ADH (**antidiuretic hormone**) increases the **permeability of the collecting duct to water**. This enables more **water reabsorption** into the bloodstream, reducing water loss. As a result, **concentrated urine** is produced, particularly during dehydration, helping the body conserve water.

MARKING SCHEME (3 Marks)
• Permeability (1), water reabsorption (1), urine concentration (1).

Q.7 What is meant by "countercurrent" in the kidney's nephron system?

Ans. The term "**countercurrent**" refers to the **opposite direction of fluid flow** in the descending and ascending limbs of the loop of Henle, as well as between the nephron and the vasa recta. This arrangement allows **efficient exchange of water and solutes**, which is crucial for establishing the osmotic gradient that enables urine concentration.

MARKING SCHEME (3 Marks)
• Definition (2), importance (1).

Q.8 How does aldosterone affect the composition of urine?

Ans. Aldosterone acts on the **distal convoluted tubule and collecting duct**, where it promotes **reabsorption of sodium ions** and **secretion of potassium ions**. As sodium reabsorption increases, water follows osmotically, leading to **water retention**. This process helps maintain **blood pressure and fluid balance**, while making urine more concentrated.

MARKING SCHEME (3 Marks)
• Functions (2), effect on urine concentration (1).

Q.9 How does the kidney regulate water balance in different hydration states?

- Ans. • During **dehydration**: ADH secretion increases, which raises the permeability of the collecting ducts. This causes more water reabsorption and the production of **concentrated urine**.
- During **overhydration**: ADH secretion decreases, lowering water reabsorption. This results in the production of **dilute urine**, preventing excess water retention.

MARKING SCHEME (3 Marks)

- Role of ADH in dehydration (1.5), role of ADH in overhydration (1.5).

Thus, the kidney regulates water balance by adjusting urine concentration according to hydration state.

Q.10 Why is the ascending limb of the loop of Henle impermeable to water, and what is the significance of this?

- Ans. The ascending limb of the loop of Henle is **impermeable to water**, but it actively transports sodium and chloride ions into the surrounding medulla. This raises the **solute concentration** of the interstitial fluid, which draws water out of the descending limb and collecting ducts. This mechanism is significant for establishing the osmotic gradient that allows the kidney to produce **concentrated urine** when needed.

MARKING SCHEME (3 Marks)

- Reason (1), effect (1), significance (1).

SLO [B-12-R-19]

List the urinary tract infections and the bacteria responsible.

URINARY TRACT INFECTIONS (UTIS) AND THEIR CAUSATIVE BACTERIA

Urinary tract infections (UTIs) are among the most frequent bacterial infections in humans, affecting millions of people worldwide each year. The urinary system, which includes the kidneys, ureters, bladder, and urethra, is normally kept sterile by various defense mechanisms such as regular urination, mucosal barriers, and immune responses. However, when bacteria gain entry and begin to multiply within this system, infection occurs. UTIs are more common in women due to their shorter urethra and its close proximity to the anal opening, which facilitates the transfer of bacteria.

From a medical perspective, UTIs can be classified according to the site of infection: urethritis (infection of the urethra), cystitis (infection of the bladder), and pyelonephritis (infection of the kidneys). In some cases, bacteria may be present in urine without obvious symptoms; this condition is called asymptomatic bacteriuria. Each type of infection has distinct clinical significance and is caused by characteristic groups of bacteria.

1. Urethritis

Urethritis refers to the infection of the urethra, the narrow tube that carries urine from the bladder to the outside. It often presents with painful urination (dysuria), itching, or discharge. While *Escherichia coli* (*E. coli*), a common bacterium of the gut, can cause urethritis, two other important bacterial agents are sexually transmitted: *Chlamydia trachomatis* and *Neisseria gonorrhoeae*. These pathogens are responsible for a large proportion of sexually acquired urethral infections in young adults. If untreated, such infections may spread to the reproductive tract, causing complications such as pelvic inflammatory disease in females or infertility in males.

2. Cystitis

When the infection spreads upward from the urethra into the bladder, it leads to cystitis. This condition is especially common in women and is characterized by urinary urgency, frequent urination in small amounts, pain during urination, and cloudy or foul-smelling urine. The **most common bacterium responsible is *Escherichia coli***, which accounts for nearly 80–85% of all bladder infections. Another bacterium, *Staphylococcus saprophyticus*, is frequently implicated in young, sexually active females. Less common agents include *Klebsiella pneumoniae* and *Proteus mirabilis*, which are often associated with hospital-acquired infections.

3. Pyelonephritis

If the infection continues to ascend, it can reach the kidneys, causing pyelonephritis. This is a serious condition marked by fever, chills, back pain (flank pain), and sometimes nausea or vomiting. *Escherichia coli* remains the **primary cause**, but other bacteria such as *Proteus mirabilis*, *Klebsiella pneumoniae*, and *Pseudomonas aeruginosa* are also involved, especially in patients with long-term catheters or structural abnormalities of the urinary tract. Pyelonephritis can lead to permanent kidney damage if not treated promptly with antibiotics.



4. Asymptomatic Bacteriuria

In some individuals, bacteria can be present in the urine without causing noticeable symptoms. This condition, known as asymptomatic bacteriuria, is most commonly caused by *Escherichia coli*, but may also involve *Enterococcus faecalis* or *Group B Streptococcus*. Although it usually does not require treatment in healthy individuals, it is clinically significant in **pregnant women**, as untreated bacteriuria may progress to kidney infections, potentially leading to complications for both the mother and the fetus.

► Major Causative Bacteria of UTIs

While different bacteria are associated with different forms of UTIs, some species stand out as the most important pathogens:

- *Escherichia coli* → the single most common cause of all types of UTIs.
- *Staphylococcus saprophyticus* → frequently causes bladder infections in young women.
- *Klebsiella pneumoniae* and *Proteus mirabilis* → cause complicated UTIs and are often linked with hospital patients.
- *Pseudomonas aeruginosa* → a dangerous pathogen in catheterized or immunocompromised patients.
- *Chlamydia trachomatis* and *Neisseria gonorrhoeae* → cause urethritis, particularly as sexually transmitted infections.
- *Enterococcus faecalis* → occasionally implicated in persistent UTIs.

Type of Infection	Primary Symptoms	Major Causative Bacteria	Notes/Remarks
Urethritis (infection of urethra)	Painful urination, burning, discharge	<i>Escherichia coli</i> , <i>Chlamydia trachomatis</i> , <i>Neisseria gonorrhoeae</i>	Often sexually transmitted (Chlamydia, Gonorrhea)
Cystitis (bladder infection)	Urgency, frequent urination, cloudy urine, pelvic pain	<i>Escherichia coli</i> (80–85%), <i>Staphylococcus saprophyticus</i> , <i>Klebsiella pneumoniae</i> , <i>Proteus mirabilis</i>	More common in women; <i>S. saprophyticus</i> often in young females
Pyelonephritis (kidney infection)	Fever, back/flank pain, chills, nausea	<i>Escherichia coli</i> , <i>Proteus mirabilis</i> , <i>Klebsiella pneumoniae</i> , <i>Pseudomonas aeruginosa</i>	Can lead to permanent kidney damage if untreated
Asymptomatic bacteriuria (bacteria in urine, no symptoms)	Usually none	<i>Escherichia coli</i> , <i>Enterococcus faecalis</i> , <i>Group B Streptococcus</i>	Clinically important in pregnancy; may lead to complications if untreated

SLO [B-12-R-20]

Explain the causes and treatments of kidney stones.

Kidney Stones (Renal Calculi)

Kidney stones, medically termed **renal calculi**, are hard crystalline deposits that form within the kidneys or urinary tract. They are a common disorder of the urinary system and can cause severe pain and complications if not treated. Stones are usually composed of minerals and salts, and their size can vary from tiny grains resembling sand to large masses that obstruct the urinary tract.

The development of kidney stones is influenced by dietary habits, dehydration, infections, genetic predisposition, and certain metabolic disorders. Understanding the **causes and treatments** is essential for both prevention and management.

► Causes of Kidney Stones

1. Dehydration

One of the primary causes of stone formation is **insufficient water intake**. When urine becomes highly concentrated due to dehydration, dissolved salts such as calcium, oxalate, and uric acid crystallize and form stones. Hot climates, excessive sweating, or neglecting adequate fluid intake increase this risk.

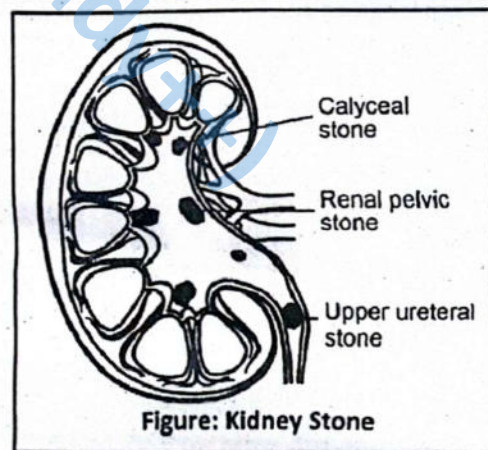


Figure: Kidney Stone

2. High Mineral and Salt Levels

- **Calcium Stones:** The most common type, often formed when calcium combines with oxalate or phosphate. Excessive dietary salt increases calcium excretion in urine, promoting stone formation.
- **Uric Acid Stones:** Develop in people with high-protein diets or conditions like gout, where uric acid levels are elevated.
- **Struvite Stones:** Associated with chronic urinary tract infections caused by bacteria that produce urease, which alters urine pH.
- **Cystine Stones:** Rare, but occur in individuals with a genetic disorder (cystinuria) that causes cystine, an amino acid, to leak into urine.

Do You Know?

The science concerned with the structure, functions and diseases of the kidneys is called nephrology.

3. Medical Conditions

Certain conditions predispose individuals to kidney stones, such as hyperparathyroidism (excess calcium release), obesity, diabetes, and chronic UTIs.

4. Family History and Genetics

A family history of stones increases susceptibility. Inherited conditions like cystinuria directly contribute to stone formation.

► Symptoms of Kidney Stones

While small stones may pass unnoticed, larger ones can cause distressing symptoms, including:

- Severe, sharp pain in the lower back or side (renal colic).
- Pain radiating to the lower abdomen or groin.
- Blood in urine (hematuria).
- Frequent and painful urination.
- Nausea and vomiting due to pain.

These symptoms often occur suddenly when a stone obstructs the flow of urine.

► Types of Urinary Stones

Urinary stones, or renal calculi, are solid mineral and salt aggregates that form within the urinary tract. Based on their chemical composition, urinary stones are classified into five major types:

- **Calcium Oxalate Stones:** These are the most common type, typically formed when urine contains high levels of calcium and oxalate. They are often hard and irregular in shape.
- **Calcium Phosphate Stones:** These stones usually develop in alkaline urine and may be associated with conditions such as renal tubular acidosis.
- **Magnesium Ammonium Phosphate Stones (Struvite Stones):** Commonly seen in individuals with recurrent urinary tract infections (UTIs), these stones form in alkaline urine and can grow rapidly. Large struvite stones may form branching structures known as **staghorn calculi**, which can occupy a major portion of the renal pelvis and calyces.
- **Uric Acid Stones:** These stones are formed in persistently acidic urine and are often linked to conditions such as gout or high purine intake. Many uric acid stones contain a combination of **uric acid and calcium oxalate**.
- **Cystine Stones:** These are rare and typically seen in individuals with a hereditary disorder called cystinuria, where cystine—a type of amino acid—accumulates in the urine.

Most urinary stones range from **2–3 mm** in diameter, although their size and surface texture (smooth or uneven) can vary depending on their composition and site of formation. Large, branching stones resembling the horns of a stag are termed **staghorn stones**, and they often require surgical intervention.

► Treatments of Kidney Stones

1. Increased Fluid Intake

For small stones, the simplest and most effective treatment is **drinking large amounts of water** (2–3 liters daily) to flush the stones naturally through urine.

2. Medications

- **Pain Relievers:** Nonsteroidal anti-inflammatory drugs (NSAIDs) help manage severe pain.
- **Alpha-blockers (e.g., tamsulosin):** Relax the ureter muscles, making it easier for stones to pass.
- **Antibiotics:** Given if stones are associated with urinary infections.



3. Medical Procedures

- **Extracorporeal Shock Wave Lithotripsy (ESWL):** High-energy sound waves break large stones into smaller fragments that pass naturally.
- **Ureteroscopy:** A thin scope inserted through the urethra and bladder into the ureter to locate and remove stones.
- **Percutaneous Nephrolithotomy:** A minimally invasive surgical procedure to remove very large or stubborn stones directly from the kidney.

4. Preventive Measures

- Maintaining proper hydration.
- Reducing salt, oxalate-rich foods (like spinach, nuts, and tea), and animal proteins in diet.
- Using prescribed medications to reduce recurrence in people prone to frequent stone formation.

SLO [B-12-R-21]

Outline the causes of kidney failure.

Kidney Failure

Renal failure refers to a decline in the kidneys' ability to filter waste and maintain homeostasis.

A. Chronic Renal Failure

- Progressive and irreversible loss of renal function, developing over months to years.
- May lead to end-stage renal disease (ESRD) requiring dialysis or transplantation.

Major Causes:

1. **Chronic bacterial infections** of the renal pelvis.
2. **Nephritis** (inflammation of glomeruli).
3. **Uncontrolled hypertension** (high blood pressure).
4. **Diabetes mellitus** (leading to diabetic nephropathy).

B. Acute Renal Failure

- Sudden loss of kidney function, often reversible if treated early.

Common Causes:

Factor	Explanation
Severe hemorrhage	Reduces blood supply to kidneys
Dehydration	Due to vomiting, diarrhea, or excessive sweating
Obstruction	Stones or tumors blocking urinary flow
Nephritis	Acute inflammation of kidney tissue

Comparison of Chronic Renal Failure and Acute Renal Failure:

Feature	Chronic Renal Failure	Acute Renal Failure
Onset	Gradual	Sudden
Reversibility	Irreversible	Reversible if treated early
Duration	Months to years	Days to weeks
Common causes	Diabetes, hypertension, nephritis	Hemorrhage, dehydration, obstruction

Q. Hypothesize kidney stone by studying the urine test of relevant patient.

- When urine is acidic (low pH) the stone is of calcium oxalate.
- When the urine is alkaline (high pH) the stone is of calcium phosphate.
- When urine is persistently acidic the stone is of uric acid type.

Ans. By analyzing the pH of the patient's urine sample, one can infer the likely type of kidney stone:

- If the urine is acidic (low pH), the patient is most likely forming calcium oxalate or uric acid stones.
- If the urine is persistently acidic, it is highly suggestive of uric acid stones.
- If the urine is alkaline (high pH), the stone is most likely composed of calcium phosphate.

SCIENCE TITBITS

Chronic kidney failure can progress to end-stage renal disease (ESRD) and uremia, which is fatal unless artificial filtering (dialysis) or a kidney transplant.



KIDNEY STONES AND KIDNEY FAILURE

- A patient with persistent acidic urine is likely to develop which type of kidney stone?
A. Calcium phosphate stone
B. Calcium oxalate stone
C. Cysteine stone
D. Uric acid stone
- Which of the following explains why urinary tract infections are more common in females than males?
A. Higher rate of bacterial exposure
B. Larger urinary bladder
C. Shorter urethra allowing easier bacterial entry
D. Hormonal differences affecting immunity
- In kidney stone patients, which of the following combinations is most commonly responsible for stone formation?
A. Hyperuricemia and acidic urine
B. Hypercalciuria and hyperoxaluria
C. Dehydration and low phosphate levels
D. Hyperkalemia and high blood pressure
- Which condition is a major cause of chronic renal failure?
A. Acute trauma
B. Vomiting and diarrhoea
C. Nephritis or diabetic complications
D. Kidney stone dislodgement
- Which statement best explains the working of Extracorporeal Shock Wave Lithotripsy (ESWL)?
A. Shock waves are generated from inside the bladder
B. Laser beams are used through a catheter
C. External shock waves break kidney stones into fragments
D. Surgical removal of stones through open incision
- A stone formed in alkaline urine is most likely composed of:
A. Calcium oxalate
B. Uric acid
C. Calcium phosphate
D. Cysteine
- Which condition most likely leads to acute renal failure?
A. Slow glomerular decline due to diabetes
B. Haemorrhage due to trauma
C. Long-term hypertension
D. Bladder infections
- What is a likely outcome of nephrolithiasis if left untreated?
A. Bladder cancer
B. Kidney regeneration
C. Obstruction and kidney failure
D. Glomerular enlargement

Answer Key

1. D 2. C 3. B 4. C 5. C 6. C 7. B 8. C

Short Questions

Q.1 What is nephrolithiasis and what causes it?

Ans. Nephrolithiasis is the medical term for the formation of kidney stones—solid crystalline deposits that develop inside the kidneys. The main causes include excess calcium in urine (hypercalciuria), high oxalate (hyperoxaluria), or increased uric acid (hyperuricemia). Dehydration is a major risk factor, as it reduces urine volume and increases mineral concentration.

MARKING SCHEME (3 Marks)

- Definition (1), causes (2).

Q.2 Urine pH influences the type of kidney stone formed. Justify.

Ans. The pH of urine plays a key role in determining which type of stone forms:

- Acidic urine favors the formation of uric acid and calcium oxalate stones.
 - Alkaline urine promotes the development of calcium phosphate stones.
- Thus, maintaining a balanced urine pH through hydration and dietary control reduces the likelihood of stone formation.

MARKING SCHEME (3 Marks)

- Stones in acidic urine (1.5), stones in alkaline urine (1.5).

Q.3 What are the main symptoms of kidney stones?

Ans. Symptoms commonly include:

- Severe flank or lower back pain (renal colic).
- Blood in urine (hematuria).

MARKING SCHEME (3 Marks)

- Six symptoms (3).



3. Nausea and vomiting.
4. Cloudy or foul-smelling urine.
5. Urinary urgency or difficulty passing urine.
6. Fever and chills (if infection occurs).

In severe cases, stones may obstruct urine flow, leading to complications.

Q.4 Describe the ESWL technique used for kidney stone treatment.

Ans. Extracorporeal Shock Wave Lithotripsy (ESWL) is a non-invasive treatment for kidney stones. It uses focused high-energy shock waves directed from outside the body to break stones into smaller fragments. These fragments can then pass naturally through the urinary tract. ESWL is usually recommended for stones less than 2 cm in size.

MARKING SCHEME (3 Marks)
• Definition (1), principle (2).

Q.5 What is the role of PCNL in stone removal?

Ans. Percutaneous Nephrolithotomy (PCNL) is a minimally invasive surgical procedure used for removing large or complex kidney stones. A small incision is made in the back to create a tract into the kidney. Through this tract, instruments and ultrasound probes are inserted to fragment and extract the stones directly.

MARKING SCHEME (3 Marks)
• Definition (1), principle (2).

Q.6 Differentiate between acute and chronic kidney failure.

- Ans.**
- **Acute kidney failure:** sudden loss of kidney function, often due to trauma, severe dehydration, shock, or obstruction. It is usually reversible if treated promptly.
 - **Chronic kidney failure:** gradual decline in kidney function over years, commonly caused by diabetes, hypertension, or chronic glomerulonephritis. It leads to irreversible damage requiring dialysis or transplantation.

MARKING SCHEME (3 Marks)
• Definition (1), causes (1), effects (1) for both types.

Q.7 What are the main causes of urinary tract infections (UTIs)?

Ans. The majority of UTIs are caused by *Escherichia coli* (*E. coli*) entering the urinary tract from the anal region. Women are more prone due to their short urethra. Other causes include poor hygiene, prolonged catheter use, incomplete bladder emptying, or obstruction. UTIs may affect the urethra (urethritis), bladder (cystitis), or kidneys (pyelonephritis).

MARKING SCHEME (3 Marks)
• Three named causes (3).

Q.8 How can kidney stones be prevented?

- Ans.** Kidney stones can be prevented by:
1. **Increasing fluid intake** (especially water) to dilute urine.
 2. **Dietary control:** reducing oxalate-rich foods (e.g., spinach), limiting salt, and moderating animal protein.
 3. **Maintaining healthy body weight** and physical activity.
 4. **Monitoring urine pH** and treating underlying metabolic conditions.

MARKING SCHEME (3 Marks)
• Any three preventive measures (3).

SLO [B-12-R-22]

Explain in detail the mechanism and problems related to dialysis.

DIALYSIS: MECHANISM

Renal dialysis is a medical procedure used to remove toxins, excess water, and waste products from the blood when the kidneys can no longer perform these vital functions adequately. Although dialysis mimics the natural function of kidneys, it is comparatively less efficient and cannot entirely replicate their complex processes. There are two primary types of dialysis: **haemodialysis** and **peritoneal dialysis**.

Haemodialysis

Haemodialysis involves the removal of waste materials and excess fluids by circulating the patient's blood through an external filtration system known as a **dialyzer**. This system uses tubes made of **semipermeable membranes** that allow the selective exchange of substances.



A catheter is inserted into a large blood vessel, usually in the arm, through which blood is withdrawn and passed into the dialysis machine. The blood pump directs the blood into the dialyzer from the top. Inside the dialyzer, blood flows through synthetic cellophane tubing, while a special fluid called **dialysate** flows around these tubes in the opposite direction. The **dialysate**, which does not contain waste products, draws out urea, creatinine, excess ions, and other metabolic wastes from the blood via **diffusion** across the semipermeable membrane.

Once filtered, the cleaned blood exits from the bottom of the machine and is returned to the patient's body through a second catheter. The used dialysate exits from the top of the machine, carrying the filtered waste materials with it.

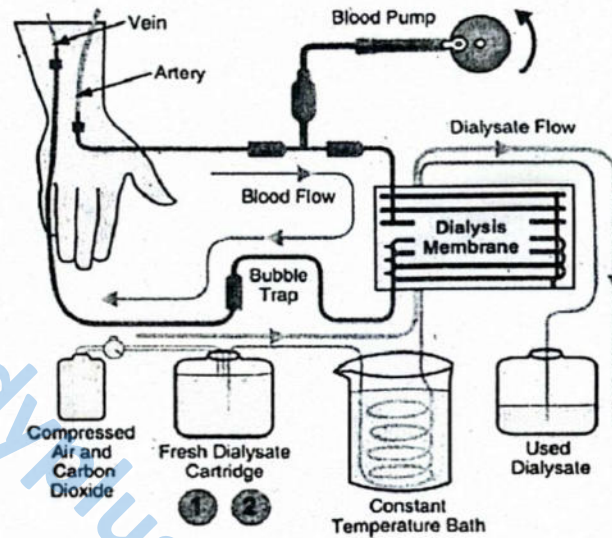


Figure: Haemodialysis

Peritoneal Dialysis

Peritoneal dialysis utilizes the body's own **peritoneal membrane**—a natural semipermeable lining of the abdominal cavity—as the filtration surface. In this method, two surgically placed catheters serve as channels for the introduction and removal of dialysate.

During the process, dialysate is infused into the **peritoneal cavity**. As blood flows through the capillaries lining the peritoneum, waste products and excess ions move across the membrane into the dialysate by **diffusion and osmosis**. After a set dwell time, the fluid containing these wastes is drained out and replaced with fresh dialysate. This exchange may be repeated several times a day, depending on the patient's clinical needs.

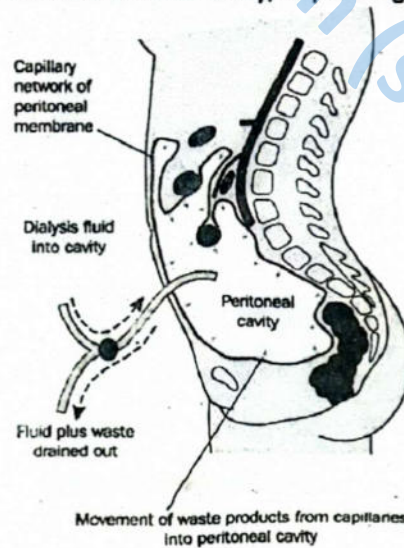


Figure: Peritoneal dialysis

Comparison of Dialysis Methods

Feature	Haemodialysis	Peritoneal Dialysis
Membrane Used	Artificial semipermeable membrane (dialyzer)	Natural peritoneal membrane
Blood Access Site	Catheter in a vein (usually in the arm)	Catheters in the abdominal cavity
Location of Procedure	Usually in a hospital or dialysis center	Can be performed at home
Blood Circulation	Outside the body	Blood remains inside the body
Frequency	Typically 2–3 sessions per week	Performed daily (multiple exchanges)
Risk of Infection	Moderate (bloodstream infection)	Higher (peritonitis if hygiene is poor)

Problems and Complications Related to Dialysis

While dialysis is life-saving, it is not a perfect substitute for kidney function and is associated with several challenges:

1. Medical Complications

- **Infections:** Hemodialysis access sites and peritoneal catheters are prone to infection. Peritoneal dialysis may cause **peritonitis**, a serious abdominal infection.
- **Hypotension:** Rapid removal of fluid during hemodialysis can cause sudden drops in blood pressure.
- **Electrolyte Imbalances:** Incorrect dialysate composition may lead to dangerous levels of potassium, sodium, or calcium.
- **Anemia and Bone Disease:** Dialysis cannot replace the kidney's hormonal functions, such as producing erythropoietin (for red blood cells) or activating vitamin D. Patients often develop anemia and weak bones.

2. Lifestyle Challenges

- **Time-Consuming:** Hemodialysis requires long sessions multiple times a week, interfering with work and daily life.
- **Dietary Restrictions:** Patients must limit intake of fluids, salt, potassium, and phosphorus.
- **Dependence on Machines:** Dialysis cannot fully restore health; patients remain dependent on treatment for survival unless they receive a kidney transplant.

3. Emotional and Financial Burden

- Dialysis treatment is costly and requires regular hospital visits.
- Long-term patients may experience psychological stress, depression, and reduced quality of life.

► Real-Life Relevance

In Pakistan and many other countries, thousands of patients rely on dialysis due to the rising incidence of diabetes and hypertension, the two major causes of kidney failure. While dialysis prolongs life, it is not a cure. Kidney transplantation remains the most effective long-term solution, but limited availability of donors and high costs make dialysis the only option for many.

Advances in portable and home-based dialysis are improving quality of life, but prevention of kidney diseases through **healthy lifestyle, proper hydration, and early treatment of diabetes and hypertension** remains the best strategy.

SLO [B-12-R-23]

Describe the principles and the problems associated with kidney transplant.

KIDNEY TRANSPLANT: PRINCIPLES AND ASSOCIATED PROBLEMS

When kidneys fail permanently and can no longer perform their vital functions of excretion and osmoregulation, patients require artificial support through dialysis or surgical replacement of the kidney. **Kidney transplantation** is the most effective long-term treatment for **end-stage renal disease (ESRD)**. It involves surgically placing a healthy kidney from a donor into a patient whose kidneys are no longer functioning. Unlike dialysis, which only partially replaces kidney function, transplantation restores most natural kidney activities, including hormone production, waste removal, and osmoregulation.



Principles of Kidney Transplant

Kidney transplantation is the most effective long-term treatment for patients suffering from end-stage renal failure. However, the success of this procedure depends on several **biological, medical, and surgical principles**. Understanding these principles helps explain why some transplants succeed for decades, while others fail within a short time.

1. Donor-Recipient Matching

The most crucial factor in a successful kidney transplant is ensuring **compatibility** between the donor and recipient.

- **Human Leukocyte Antigen (HLA) Matching:** The HLA system is a group of proteins found on the surface of white blood cells and other tissues. They act like "biological ID cards" that help the immune system distinguish between self and non-self. A closer HLA match between donor and recipient reduces the risk of the body rejecting the transplanted kidney. For example, siblings often make better donors because they share more HLA genes.
- **Blood Group Compatibility:** Matching the **ABO blood group system** (A, B, AB, O) between donor and recipient is equally important. A mismatch in blood type can trigger immediate rejection (hyperacute rejection), making transplantation impossible.

Careful donor-recipient matching is therefore the foundation of transplant success.

2. Source of Donor Kidneys

Kidneys for transplantation can come from either **living donors** or **deceased donors**.

- **Living Donors:** Often, close relatives such as parents, siblings, or even children donate a kidney. Living donor kidneys are usually healthier and function more effectively after transplantation. They also allow the surgery to be planned in advance, minimizing waiting time. Importantly, humans can live normally with one kidney, making living donation safe in most cases.
- **Deceased Donors:** These are individuals who have suffered brain death or cardiac death but whose kidneys are still healthy. Such kidneys can be transplanted if the individual consented to organ donation during their lifetime, or if their family permits it after death. Deceased donor programs expand the pool of available kidneys but outcomes may sometimes be slightly less favorable than with living donors.

In countries like Pakistan, deceased donor programs are limited, which adds to the shortage of kidneys and prolongs patient waiting times.

3. Surgical Procedure

The surgical placement of a transplanted kidney follows specific principles to ensure both function and safety.

- **Placement of the Kidney:** The new kidney is placed in the **lower abdomen (iliac fossa)**, near the pelvis, rather than in the original position of the natural kidneys. This location provides easier access to blood vessels and the bladder.
- **Connection of Blood Vessels:** The donor kidney's **renal artery and renal vein** are surgically attached to the recipient's **iliac artery and iliac vein**, ensuring proper blood flow.
- **Connection of the Ureter:** The donor kidney's ureter is connected directly to the recipient's bladder, allowing urine to pass out normally.
- **Retention of Diseased Kidneys:** In most cases, the patient's damaged kidneys are **not removed**. They are left in place unless they cause complications such as recurrent infections, severe hypertension, or kidney cancers.

This surgical strategy minimizes risks and ensures the transplanted kidney can function quickly after blood flow is restored.

4. Immunosuppression

Since the transplanted kidney comes from another individual, the recipient's immune system naturally recognizes it as **foreign**. Without intervention, the immune system would attack the graft, leading to rejection. To prevent this, patients must take **immunosuppressive drugs for life**.

- **Common Immunosuppressants:** Drugs such as **cyclosporine, tacrolimus, azathioprine, and corticosteroids** are given in combination. These reduce the activity of immune cells that would otherwise **damage the graft**.
- **Balancing Risks:** While these drugs are essential for graft survival, they also weaken the patient's defense against infections and cancers. Doctors must carefully adjust doses to balance the risk of rejection with the side effects of long-term immunosuppression.

Without immunosuppressive therapy, no kidney transplant could succeed.



Problems Associated with Kidney Transplant

Kidney transplantation is considered the gold standard treatment for end-stage renal failure because it restores natural kidney function more effectively than dialysis. However, despite its advantages, it is not free from challenges. Medical, surgical, and social complications can affect the success of the procedure and the patient's long-term survival. The following are the major problems associated with kidney transplantation:

1. Organ Rejection

The immune system is designed to protect the body from foreign invaders. Unfortunately, a transplanted kidney is also recognized as "foreign," even if it comes from a close relative. This immune attack is known as rejection.

- **Hyperacute Rejection:** This occurs within minutes to hours after transplantation if the donor and recipient tissues are poorly matched. The patient's antibodies attack the kidney immediately, causing clotting within the blood vessels of the graft. Hyperacute rejection is rare today because careful tissue and blood matching is done before surgery.
- **Acute Rejection:** This typically develops days to weeks after the operation. It is caused by the recipient's immune cells (T lymphocytes) attacking the kidney tissue. Modern immunosuppressive drugs can often reverse acute rejection if detected early.
- **Chronic Rejection:** This is a slow, progressive rejection that occurs months or years after transplantation. The blood vessels of the transplanted kidney gradually thicken and scar tissue forms, leading to a steady decline in kidney function. Unlike acute rejection, chronic rejection is usually irreversible and is the leading cause of long-term graft failure.

Even with strong immunosuppressive therapy, rejection remains the most serious complication of kidney transplantation and requires lifelong monitoring.

2. Side Effects of Immunosuppressive Drugs

To prevent rejection, patients must take immunosuppressive drugs for life. While these medicines are lifesaving, they also create new health challenges:

- **Weak Immune Defense:** Immunosuppressants reduce the body's ability to fight infections. Patients are more prone to bacterial, viral, and fungal infections, including pneumonia, urinary tract infections, and shingles. Even minor infections can become life-threatening.
- **Delayed Healing:** Surgical wounds and minor injuries take longer to heal because of weakened immune activity.
- **Cancer Risk:** Prolonged suppression of immunity increases the likelihood of certain cancers, especially skin cancers and lymphomas.
- **Metabolic Side Effects:** Drugs such as corticosteroids and calcineurin inhibitors may cause high blood pressure, diabetes, osteoporosis (bone weakness), and excessive hair growth.
- Thus, while immunosuppressive drugs protect the graft, they also create a delicate balance between preventing rejection and maintaining overall health.

3. Shortage of Donor Kidneys

The demand for kidney transplantation far exceeds the supply of donor organs worldwide.

- **Long Waiting Lists:** Thousands of patients remain dependent on dialysis for years, waiting for a compatible kidney. In many countries, the waiting period can stretch from 3 to 7 years.
- **Limited Donor Sources:** Living donors (usually relatives) are more successful, but not always available. Deceased donor programs exist, but cultural beliefs and lack of awareness about organ donation often limit participation.
- **Ethical and Social Concerns:** The shortage has led to unethical practices such as illegal organ trade and exploitation of the poor, especially in developing countries. International guidelines now promote strict regulation of organ donation to prevent exploitation.

This shortage remains a major barrier to successful transplantation programs, particularly in countries like Pakistan where awareness about cadaveric donation is still low.

4. Surgical and Medical Complications

As with any major surgery, kidney transplantation carries inherent risks, both during and after the procedure.

- **Intraoperative Risks:** Complications during surgery may include bleeding, blood clots, or injury to surrounding structures. These are managed with careful surgical techniques and monitoring.
- **Postoperative Infections:** Surgical wounds and urinary tract infections are common due to immunosuppression.



- **Delayed Graft Function (DGF):** In some cases, the transplanted kidney does not start working immediately. The patient may need dialysis in the first few days or weeks until the new kidney recovers.
- **Recurrence of Original Kidney Disease:** In certain conditions (e.g., glomerulonephritis), the disease that damaged the original kidneys can also affect the transplanted one.

Kidney transplantation has dramatically improved survival and quality of life for patients with ESRD. In Pakistan and many other countries, however, limited availability of donors, high costs, and lack of awareness about organ donation programs restrict access to this treatment. Patients often rely on living related donors, which raises ethical and medical challenges.

Modern advances such as **paired kidney exchange programs** (where families swap donors for better compatibility) and **xenotransplantation research** (using genetically modified animal organs) may help address the shortage in the future.

SLOs Based (MCQs)

DIALYSIS AND KIDNEY TRANSPLANT

- Which structure is used as a semipermeable membrane in haemodialysis?
A. Cellulose sheet B. Capillary endothelium
C. Cellophane tube D. Dialysate bag
- In peritoneal dialysis, which natural membrane serves the role of filtration?
A. Renal capsule B. Peritoneum
C. Glomerulus D. Ureter
- Which of the following statements is true regarding haemodialysis?
A. Blood is filtered inside the body through the liver
B. Dialysate and blood flow in the same direction
C. Blood circulates outside the body through a dialyzer
D. No membrane is involved in the process
- Which condition most commonly necessitates kidney transplantation?
A. Gallstones B. Acute appendicitis
C. Chronic kidney failure D. Bladder infection
- Which of the following must be matched between a kidney donor and recipient?
A. Platelet count
B. Blood glucose level
C. ABO blood group and HLA
D. Hemoglobin level
- Cyclosporine is given to kidney transplant patients to:
A. Enhance urine output
B. Stimulate erythropoietin production
C. Suppress the immune response
D. Prevent blood clotting
- The risk of rejection in kidney transplant increases when the patient:
A. Eats protein-rich food
B. Misses immunosuppressive medication
C. Drinks excess water
D. Exercises too much

Answer Key

1. C 2. B 3. C 4. C 5. C 6. C 7. B

Short Questions

Q.1 What is haemodialysis and how does it work?

Ans. Haemodialysis is an artificial method of filtering blood when the kidneys fail. The patient's blood is circulated through a **dialyzer**, which contains **cellophane tubes acting as a semipermeable membrane**. Waste products and excess water diffuse out into the dialysis fluid, and the **purified blood is returned** to the patient's body.

MARKING SCHEME (3 Marks)

- Definition (1), working principle (2).

Q.2 What is kidney transplantation and when is it required?

Ans. Kidney transplantation is a **surgical procedure** in which a healthy kidney from a donor is implanted into a patient. It is required in cases of **chronic or end-stage kidney failure**, where kidneys have permanently lost their filtering function. Compared to dialysis, transplantation offers **better long-term survival and quality of life**.

MARKING SCHEME (3 Marks)

- Definition (1), reasons for transplant (2).

Q.3 What are the key requirements for a successful kidney transplant?

Ans. A successful kidney transplant requires:



1. ABO blood group compatibility between donor and recipient.
2. HLA (human leukocyte antigen) matching to reduce immune rejection risk.
3. A suitable donor (living or deceased), since one healthy kidney is sufficient for normal life.

MARKING SCHEME (3 Marks)
• 1 mark each for three requirements.

Q.4 Why are immunosuppressive drugs necessary after a kidney transplant?

Ans. Immunosuppressive drugs (e.g., cyclosporine) are necessary to prevent the immune system from rejecting the transplanted kidney. These drugs suppress immune responses, allowing the graft to function. Patients must take them lifelong, as stopping medication can lead to late rejection even years after surgery.

MARKING SCHEME (3 Marks)
• Importance (2), precaution (1).

Q.5 What complications are associated with kidney transplantation?

- Ans.**
1. **Graft rejection** – the immune system may recognize the transplanted kidney as foreign and attack it.
 2. **Drug toxicity** – immunosuppressive drugs may cause side effects such as infections, liver damage, or high blood pressure.
 3. **Ongoing medical monitoring** is essential to balance both risks.

MARKING SCHEME (3 Marks)
• Graft rejection (1.5), drug toxicity (1.5).

Exercise

I

Multiple Choice Questions (MCQs)

❖ Select the correct Answer.

1. Excretion of hypotonic urine in humans is associated best with the:
A. glomerular capsule
B. proximal convoluted tubule
C. loop of the Henle
D. distal convoluted tubule
2. The walls of the _____ are made more or less permeable to water, depending on the need to conserve water:
A. ureter
B. urethra
C. fibrous capsule
D. collecting duct
3. Which of the following will cause a decrease in ADH production?
A. dehydration
B. an increase in osmotic pressure of blood
C. drinking water
D. abnormally low blood pressure
4. The function of glomerulus and Bowman's capsule of the nephron is to:
A. reabsorb water into the blood
B. eliminate ammonia from the body
C. reabsorb salts and amino acids
D. filter the blood and capture the filtrate
5. In man, glucose is present in blood plasma but not in urine. This is because glucose molecules are:
A. actively transported from the proximal convoluted tubule to blood capillaries
B. oxidised to supply energy for ultrafiltration
C. stored in the kidney
D. too large to enter Bowman's capsule
6. Evidence for glomerular filtration in the kidney could be obtained by comparing the sizes of the molecules present in Bowman's capsule with those in the:
A. afferent blood vessel
B. collecting duct
C. loop of Henle
D. proximal tubule
7. The site and principal mechanism for the passage of glucose into the bloodstream in the human kidney is the
A. collecting duct, by active secretion
B. glomerulus, by selective reabsorption
C. glomerulus, by ultrafiltration
D. proximal convoluted tubule, by selective reabsorption
8. A drug reduces mitochondrial activity in kidney nephrons. Which chemical will be present in increased amounts in the urine?
A. ammonia
B. glucose
C. uric acid
D. urea



9. The water content of human blood is regulated by ADH. In which part of the nephron does regulation occur?
 A. ascending limb of loop of Henle
 B. descending limb of the loop of Henle
 C. Bowman's capsule
 D. proximal convoluted tubule
10. The reason why the right kidney is slightly lower than the left is:
 A. the left kidney is bigger than right
 B. considerable space occupied by the heart
 C. considerable space occupied by the liver on the right side
 D. the right kidney is bigger than the left
11. Which of the following is known as the functional unit of the kidney and is found in the cortex?
 A. calyces
 B. renal pyramids
 C. nephrons
 D. medulla
12. This artery passes blood to the kidney:
 A. common iliac
 B. cystic
 C. renal
 D. coeliac
13. Which structure carries urine from the kidneys to the bladder?
 A. nephron
 B. ureter
 C. bladder
 D. urethra
14. The blood vessel which carries blood to the glomerulus is the:
 A. renal artery
 B. afferent arteriole
 C. renal vein
 D. peritubular capillaries
15. The basic unit of the renal system is the:
 A. cortex
 B. neuron
 C. nephron
 D. medulla
16. The centre of the kidney where urine collects before leaving the kidneys is the:
 A. glomerulus
 B. proximal tubule
 C. distal tubule
 D. renal pelvis
17. An organ or structure that is not a component of the urinary system is the:
 A. urethra
 B. urinary bladder
 C. ureter
 D. adrenal gland
18. All of the following belong to the urinary system except:
 A. urethra
 B. ureter
 C. bladder
 D. prostate
19. What portion of the nephron extends into the medulla?
 A. nephron loop
 B. proximal convoluted tubule
 C. distal convoluted tubule
 D. papillary duct
20. Fibrous connective tissue that surrounds each kidney is the:
 A. cortex
 B. hilum
 C. medulla
 D. renal capsule

Answer Key

1. C	2. D	3. C	4. D	5. A	6. A	7. D	8. B	9. D	10. C
11. C	12. C	13. B	14. B	15. C	16. D	17. D	18. D	19. A	20. D

II

Short Answer Questions

Q.1 Name the organs of the urinary system and write their major functions.

Ans. The urinary system is composed of four main organs: the **kidneys**, **ureters**, **urinary bladder**, and **urethra**. The **kidneys** are paired bean-shaped organs located in the abdominal cavity that filter the blood, remove nitrogenous wastes such as urea and creatinine, regulate pH, and maintain water-salt balance. From each kidney arises a **ureter**, a muscular tube that transports urine to the **urinary bladder**, a hollow muscular organ that temporarily stores urine until it is convenient to expel it. Finally, urine exits the body through the **urethra**, a tube that functions in micturition. Together, these organs maintain homeostasis by excreting wastes and regulating the internal environment of the body.

MARKING SCHEME (3 Marks)

- Names of organs (1.5), major functions (1.5).

Q.2 Describe glomerular filtration.

Ans. Glomerular filtration is the initial step in urine formation and occurs in the **renal corpuscle of the nephron**. Blood enters the glomerulus through the afferent arteriole under relatively high pressure. This hydrostatic pressure forces **water, glucose, salts, amino acids**, and

MARKING SCHEME (3 Marks)

- Definition (1.25), site of occurrence (0.75), substances filtered/retained (1).



nitrogenous wastes through the filtration membrane of the glomerular capillaries into the Bowman's capsule. Larger molecules such as **proteins** and **blood cells** are too big to pass through the pores, so they remain in the bloodstream. The fluid that enters the capsule is called **glomerular filtrate**, which resembles blood plasma but is protein-free. This process ensures that wastes and excess solutes are removed from blood while essential macromolecules are retained.

Q.3 Describe the counter current multiplier mechanism.

Ans. The countercurrent multiplier mechanism occurs in the **loop of Henle** and is vital for the concentration of urine. The **descending limb** of the loop is permeable to water but not to solutes. As the filtrate descends, water moves out into the surrounding hyperosmotic medulla, concentrating the filtrate. In contrast, the **ascending limb** is impermeable to water but actively transports sodium and chloride ions into the interstitial fluid. This dual action creates a steep **osmotic gradient** in the medulla, a process called the countercurrent multiplier effect. This gradient allows the collecting ducts to reabsorb more water under the influence of ADH, ensuring that urine can be concentrated when the body needs to conserve water.

MARKING SCHEME (3 Marks)

- Descending limb role (1), ascending limb role (1), significance (1).

Q.4 Name the parts of a nephron and trace the blood supply to the nephron.

Ans. A nephron, the functional unit of the kidney, consists of two main regions: the **renal corpuscle** and the **renal tubule**. The renal corpuscle includes the **glomerulus** and **Bowman's capsule**, while the tubule is divided into the **proximal convoluted tubule (PCT)**, **loop of Henle**, **distal convoluted tubule (DCT)**, and **collecting duct**. Blood reaches the nephron via the **renal artery**, which branches into **afferent arterioles** supplying each glomerulus. After filtration, blood exits through the **efferent arteriole**, which then forms either the **peritubular capillaries** around cortical nephrons or the **vasa recta** surrounding juxtamedullary nephrons. Finally, the blood is drained into the **renal vein**, returning to circulation. This close blood supply ensures efficient exchange during urine formation.

MARKING SCHEME (3 Marks)

- Nephron parts (1.5), blood supply sequence (1.5).

Q.5 Name general processes which are involved with urine formation.

Ans. Urine formation involves **three essential processes** that together ensure waste removal and conservation of useful substances.

1. **Glomerular filtration** – occurs in the renal corpuscle, where blood plasma is filtered into the Bowman's capsule under pressure, forming filtrate.
2. **Tubular reabsorption** – in the PCT, loop of Henle, DCT, and collecting duct, essential substances such as glucose, amino acids, salts, and most water are reabsorbed back into the blood through active and passive mechanisms.
3. **Tubular secretion** – additional wastes, ions (like H^+ and K^+), and drugs are actively secreted from peritubular capillaries into the nephron tubules.
4. These processes work together to maintain homeostasis by removing toxic products while conserving water and nutrients.

MARKING SCHEME (3 Marks)

- Glomerular filtration (1), reabsorption (1), secretion (1).

Q.6 Describe urinary tract infection.

Ans. Urinary tract infection (UTI) is a bacterial infection of any part of the urinary system such as urethra, bladder, or kidneys. It occurs when bacteria enter and multiply in the urinary tract. UTIs are more common in women because of their shorter urethra and proximity to the anal opening.

MARKING SCHEME (3 Marks)

- Definition of UTI (1)
- Sites of infection (1)
- Cause or prevalence in women (1)

Q.7 Name the three urinary tract infections and the bacteria responsible.

Ans. **Urethritis** – caused by *Escherichia coli*, *Chlamydia trachomatis*, *Neisseria gonorrhoeae*.

- **Cystitis** – mainly by *Escherichia coli* (80–85%), *Staphylococcus saprophyticus*, *Klebsiella pneumoniae*, *Proteus mirabilis*.
- **Pyelonephritis** – caused by *Escherichia coli*, *Proteus mirabilis*, *Klebsiella pneumoniae*, *Pseudomonas aeruginosa*.

MARKING SCHEME (3 Marks)

- Correctly naming three UTIs (1)
- Correctly linking bacteria to each (2)



Q.8 What are the causes of kidney failure?

Ans. Kidney failure may result from:

1. **Uncontrolled diabetes and high blood pressure**, which damage nephrons.
2. **Infections or blockages** like repeated UTIs or kidney stones.
3. **Exposure to toxins or drugs** that impair kidney function.

MARKING SCHEME (3 Marks)

- Any three valid causes with explanation (1 mark each).

Q.9 By what physical processes do solutes enter or leave the blood during dialysis?

Ans. In dialysis, solutes move by **diffusion**, where wastes like urea pass from blood into dialysate. Some ions move by **osmosis**, depending on concentration gradients. Larger molecules remain in the blood because the membrane is selectively permeable.

MARKING SCHEME (3 Marks)

- Diffusion explained (1), Osmosis explained (1) & Selective permeability role (1)

Q.10 Why do blood and dialysate flow in opposite direction?

Ans. Blood and dialysate flow in opposite directions to maintain a **concentration gradient** along the entire dialysis membrane. This ensures continuous removal of waste and efficient exchange of solutes. It allows maximum clearance of toxins from the blood.

MARKING SCHEME (3 Marks)

- Opposite flow mentioned (1)
- Concentration gradient maintained (1)
- Efficiency of waste removal explained (1)

Q.11 Suggest two problems that might occur if the dialysate was pure water.

Ans. Using pure water as dialysate in haemodialysis would be extremely dangerous because it would create serious osmotic and electrolyte imbalances. Firstly, since the solute concentration of pure water is essentially zero, an osmotic gradient would drive water rapidly into the patient's blood. This excessive influx could lead to **hemolysis** (bursting of red blood cells), dilution of plasma, and fluid overload in tissues, potentially causing swelling or pulmonary edema. Secondly, the absence of essential solutes in the dialysate would cause vital electrolytes such as **sodium, potassium, calcium, and chloride** to diffuse out of the blood into the dialysate. This would lead to severe **electrolyte imbalance**, disrupting nerve transmission, heart rhythm, and muscle function. Therefore, dialysate must always be carefully balanced to mimic plasma composition.

MARKING SCHEME (3 Marks)

- Osmotic imbalance (1.5), electrolyte loss (1.5).

Q.12 Why are women more likely to acquire UTI as compared to men?

Ans. Women are at a higher risk of developing urinary tract infections (UTIs) due to specific anatomical and physiological factors. The **female urethra is much shorter** (around 4 cm) compared to the male urethra (about 20 cm). This reduced length provides bacteria with a shorter distance to travel in order to reach the bladder, making infections more likely. Additionally, the **urethral opening in women is located closer to the anus**, where bacteria such as *Escherichia coli* are abundant. This close proximity increases the chance of bacterial contamination of the urethra. Other contributing factors include hormonal influences, sexual activity, and improper hygiene practices, which further predispose women to recurrent infections. These differences explain why UTIs are significantly more common in females than in males.

MARKING SCHEME (3 Marks)

- Short urethra (1), proximity to anus (1), other contributing factors (1).

Q.13 Write the differentiate between:

(a) Osmoregulation and Osmoconformers.

Ans. Osmoregulation refers to the active physiological processes by which organisms regulate water and solute concentrations in their body fluids, maintaining a relatively constant internal environment despite external changes. Humans and freshwater fishes are classic osmoregulators, as they actively excrete or conserve salts and water using kidneys, gills, and other mechanisms. In contrast, osmoconformers are organisms, mostly marine invertebrates, whose internal osmotic pressure changes with the external environment. They do not spend energy on active regulation but allow their body fluids to match seawater osmolarity. This strategy saves energy but limits survival to stable habitats. Thus, osmoregulators achieve homeostasis at an energetic cost, while osmoconformers adapt passively.

MARKING SCHEME (3 Marks)

- Osmoregulators definition + example (1.5), osmoconformers definition + example (1.5).



(b) Ammonotelic and Ureotelic Organisms.

Ans. Ammonotelic organisms excrete nitrogenous waste directly as ammonia, which is highly toxic but very soluble in water. Since ammonia must be eliminated quickly, ammonotelism is seen in aquatic organisms such as bony fishes, amphibians, and aquatic invertebrates. In contrast, ureotelic organisms excrete urea, a less toxic compound formed in the liver through the urea cycle. Urea requires more energy to produce but is safer for storage and excretion, as it requires less water. Mammals, amphibians, and some marine fishes are ureotelic. The key difference lies in water availability and habitat: ammonotelic organisms live in water-rich environments, whereas ureotelic animals are adapted to conserve water.

MARKING SCHEME (3 Marks)
• Ammonotelic (definition + example) 1.5,
ureotelic (definition + example) 1.5.

(c) Ureotelic and Uricotelic Organisms.

Ans. Ureotelic organisms excrete nitrogenous wastes primarily as urea, which is moderately soluble and less toxic than ammonia. This strategy balances water conservation with energy use, as seen in mammals, amphibians, and some marine fishes. On the other hand, uricotelic organisms excrete uric acid, an almost insoluble compound that precipitates out as a paste or crystals. This adaptation minimizes water loss, which is crucial for survival in arid environments. Birds, reptiles, and insects are classic uricotelic animals. Producing uric acid requires higher energy investment compared to urea, but it provides a significant advantage in water conservation.

MARKING SCHEME (3 Marks)
• Ureotelic (definition + example) 1.5,
uricotelic (definition + example) 1.5.

(d) Proximal and Distal Convolted Tubule.

Ans. The proximal convoluted tubule (PCT) is the first tubular structure of the nephron after Bowman's capsule. It is responsible for the bulk of reabsorption, reclaiming about 65–70% of water, sodium, chloride, glucose, amino acids, and bicarbonates. The PCT has a brush border with microvilli, which increases surface area for absorption. In contrast, the distal convoluted tubule (DCT) lies further along the nephron, after the loop of Henle. Its main function is selective reabsorption and secretion, particularly under hormonal control. For example, aldosterone enhances sodium reabsorption and potassium secretion, while parathyroid hormone regulates calcium. Unlike the PCT, the DCT has fewer microvilli and specializes in fine-tuning urine composition.

MARKING SCHEME (3 Marks)
• PCT (location + function) 1.5, DCT
(location + function) 1.5.

(e) Afferent and Efferent Arterioles.

Ans. The afferent arteriole is the blood vessel that delivers blood into the glomerulus of the nephron. It has a larger diameter compared to the efferent arteriole, creating high hydrostatic pressure that drives ultrafiltration in Bowman's capsule. The efferent arteriole, on the other hand, carries blood away from the glomerulus. Since it is narrower, it maintains pressure in the glomerular capillaries but reduces blood volume, which helps in efficient filtration. Furthermore, efferent arterioles give rise to peritubular capillaries and vasa recta that are crucial for reabsorption and concentration of urine. In summary, the afferent arteriole regulates entry into the glomerulus, while the efferent controls outflow and maintains pressure balance.

MARKING SCHEME (3 Marks)
• Afferent (definition + function) 1.5,
efferent (definition + function) 1.5.

(f) Hypercalcemia and Hyperuricemia.

Ans. Hypercalcemia refers to an abnormally high concentration of calcium in the blood, typically above 10.5 mg/dL. It may be caused by hyperparathyroidism, excessive vitamin D intake, bone diseases, or certain cancers. Symptoms include kidney stones, bone pain, abdominal discomfort, and irregular heartbeat. In contrast, hyperuricemia is defined as elevated levels of uric acid in the blood, usually above 7 mg/dL. It results from either increased production or reduced excretion of uric acid and is commonly associated with gout, kidney stones, and metabolic disorders. While hypercalcemia mainly affects calcium metabolism and bones, hyperuricemia relates to purine metabolism and can trigger painful joint inflammation.

MARKING SCHEME (3 Marks)
• Hypercalcemia (definition + cause +
symptom) 1.5, hyperuricemia
(definition + cause + symptom) 1.5.

(g) Extracorporeal Shock Wave Lithotripsy (ESWL) and Percutaneous Nephrolithotomy (PCNL).

Ans. Extracorporeal shock wave lithotripsy (ESWL) is a non-invasive treatment for kidney stones, in which high-energy sound waves are focused externally onto the stone, breaking it into small fragments that pass



naturally through urine. It is best suited for stones smaller than 2 cm. On the other hand, percutaneous nephrolithotomy (PCNL) is a minimally invasive surgical technique used for large, complex, or hard-to-reach stones. In PCNL, a small incision is made in the patient's back to access the kidney, and instruments are inserted to directly break and remove stones. While ESWL is safer and less invasive, PCNL is more effective for large stone burdens.

- MARKING SCHEME (3 Marks)**
- ESWL (definition + method) 1.5, PCNL (definition + method) 1.5.

(h) Chronic Renal Failure and Acute Renal Failure.

Ans. Acute renal failure, also called acute kidney injury (AKI), is a sudden loss of kidney function over hours or days. It is usually caused by trauma, blood loss, dehydration, obstruction, or toxins, and can often be reversed with timely treatment. Symptoms include reduced urine output, fluid retention, and electrolyte imbalance. Chronic renal failure (CRF), however, develops gradually over months or years due to long-standing conditions like diabetes mellitus, hypertension, or chronic glomerulonephritis. The damage is irreversible, progressively leading to end-stage renal disease. Symptoms include persistent fatigue, anemia, bone weakness, and accumulation of wastes. Thus, acute failure is abrupt and potentially reversible, whereas chronic failure is gradual and permanent.

- MARKING SCHEME (3 Marks)**
- Acute (definition + cause + effect) 1.5, chronic (definition + cause + effect) 1.5.

(i) Peritoneal Dialysis and Haemodialysis.

Ans. Haemodialysis uses an artificial dialyzer machine to filter wastes and excess fluid from the blood. Blood is withdrawn from the patient, passed through a semipermeable membrane in the dialyzer, and then returned. It requires regular hospital visits and specialized equipment. In contrast, peritoneal dialysis uses the patient's own peritoneal membrane (lining of the abdominal cavity) as the natural filter. A cleansing dialysate fluid is introduced into the abdominal cavity, where waste products diffuse across the peritoneal membrane, and then drained out after several hours. Peritoneal dialysis can be performed at home and offers more independence, but carries risks of infection (peritonitis).

- MARKING SCHEME (3 Marks)**
- Haemodialysis (definition + method) 1.5, peritoneal dialysis (definition + method) 1.5.

(j) Renal Cortex and Renal Medulla.

Ans. The renal cortex is the outer region of the kidney, appearing granular due to the presence of renal corpuscles (glomeruli and Bowman's capsules) and convoluted tubules. It is the primary site of filtration and much of tubular reabsorption. The renal medulla, by contrast, is the inner region, consisting mainly of renal pyramids and loops of Henle. It appears striated due to parallel arrangement of tubules and blood vessels. The medulla plays a vital role in concentrating urine through the countercurrent multiplier system and in water reabsorption under antidiuretic hormone (ADH) control. Thus, the cortex initiates urine formation, while the medulla fine-tunes its concentration.

- MARKING SCHEME (3 Marks)**
- Cortex (structure + function) 1.5, medulla (structure + function) 1.5.

(k) Vasodilation and Vasoconstriction.

Ans. Vasodilation is the widening of blood vessels due to relaxation of smooth muscles in the vessel wall, especially in arterioles. This decreases vascular resistance, lowers blood pressure, and increases blood flow to tissues. It is promoted by factors such as nitric oxide, low oxygen levels, or high carbon dioxide. In contrast, vasoconstriction is the narrowing of blood vessels due to contraction of smooth muscles. This increases resistance, raises blood pressure, and diverts blood away from less active tissues. It is triggered by sympathetic stimulation and hormones such as norepinephrine. Together, vasodilation and vasoconstriction regulate tissue perfusion, thermoregulation, and blood pressure homeostasis.

- MARKING SCHEME (3 Marks)**
- Vasodilation (definition + effect) 1.5, vasoconstriction (definition + effect) 1.5.

(l) Dialyzer and Dialysate.

Ans. A dialyzer is the artificial kidney device used in haemodialysis to filter blood. It contains thousands of fine cellophane or synthetic fibers acting as a semipermeable membrane. Blood flows on one side of the membrane, while dialysate flows on the other, allowing waste products and excess ions to pass out of the blood. Dialysate, in contrast, is the specially prepared fluid that surrounds

- MARKING SCHEME (3 Marks)**
- Dialyzer (definition + role) 1.5, dialysate (definition + role) 1.5.



the dialyzer membrane. It contains controlled concentrations of salts, glucose, and bicarbonate to mimic plasma and create gradients for diffusion. Dialysate removes urea, creatinine, and extra electrolytes while maintaining proper pH and osmotic balance. Thus, the dialyzer provides the surface, while dialysate provides the medium for exchange.



Long Answer Questions

Q.1 Discuss relationship between excretory products and habitats.

Ans. Please See SLO [B-12-R-10]

Q.2 Describe the structure and function of human kidney.

Ans. Please See SLO [B-12-R-12] & [B-12-R-13]

Q.3 Describe the structure of human nephron.

Ans. Please See SLO [B-12-R-14]

Q.4 Discuss the 'urine formation' and mechanism of urine concentration in man.

Ans. Please See SLO [B-12-R-16]

Q.5 What are kidney stones? Discuss the causes and treatment of kidney stones?

Ans. Please See SLO [B-12-R-20]

Q.6 What is renal dialysis? Describe the two types of renal dialysis.

Ans. Please See SLO [B-12-R-22]

Q.7 What is kidney transplantation? Describe principles of kidney transplant. What are the problems associated with kidney transplant?

Ans. Please See SLO [B-12-R-23]

